

RWTH AACHEN UNIVERSITY
Chair of Computer Science 5
Information Systems & Databases
Prof. Dr. Stefan Decker

Bachelor Thesis

**WILLM: A System for Academic Writing Improvement based
on Large Language Models**

Ben Thillen
Matr.-No.: 409574
Study Program: Bachelor Computer Science
October 7, 2024

Supervisors: Prof. Dr. Stefan Decker
Chair of Information Systems & Databases
RWTH Aachen University

Prof. Dr. Bernhard Rumpe
Chair of Software Engineering
RWTH Aachen University

Advisors: Yongli Mou
Chair of Information Systems & Databases
RWTH Aachen University

Fatemeh Fathi
Chair of Information Systems & Databases
RWTH Aachen University

Eidesstattliche Versicherung

Statutory Declaration in Lieu of an Oath

Thillen, Ben
Name, Vorname/Last Name, First Name

409574
Matrikelnummer (freiwillige Angabe)
Matriculation No. (optional)

Ich versichere hiermit an Eides Statt, dass ich die vorliegende ~~Arbeit~~/Bachelorarbeit/
Masterarbeit* mit dem Titel

I hereby declare in lieu of an oath that I have completed the present paper/Bachelor thesis/Master thesis* entitled

WILLM: A System for Academic Writing Improvement based on Large Language Models

selbstständig und ohne unzulässige fremde Hilfe (insbes. akademisches Ghostwriting) erbracht habe. Ich habe keine anderen als die angegebenen Quellen und Hilfsmittel benutzt. Für den Fall, dass die Arbeit zusätzlich auf einem Datenträger eingereicht wird, erkläre ich, dass die schriftliche und die elektronische Form vollständig übereinstimmen. Die Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen.

independently and without illegitimate assistance from third parties (such as academic ghostwriters). I have used no other than the specified sources and aids. In case that the thesis is additionally submitted in an electronic format, I declare that the written and electronic versions are fully identical. The thesis has not been submitted to any examination body in this, or similar, form.

Aachen, October 7, 2024
Ort, Datum/City, Date

Unterschrift/Signature

*Nichtzutreffendes bitte streichen

*Please delete as appropriate

Belehrung:

Official Notification:

§ 156 StGB: Falsche Versicherung an Eides Statt

Wer vor einer zur Abnahme einer Versicherung an Eides Statt zuständigen Behörde eine solche Versicherung falsch abgibt oder unter Berufung auf eine solche Versicherung falsch aussagt, wird mit Freiheitsstrafe bis zu drei Jahren oder mit Geldstrafe bestraft.

Para. 156 StGB (German Criminal Code): False Statutory Declarations

Whoever before a public authority competent to administer statutory declarations falsely makes such a declaration or falsely testifies while referring to such a declaration shall be liable to imprisonment not exceeding three years or a fine.

§ 161 StGB: Fahrlässiger Falscheid; fahrlässige falsche Versicherung an Eides Statt

(1) Wenn eine der in den §§ 154 bis 156 bezeichneten Handlungen aus Fahrlässigkeit begangen worden ist, so tritt Freiheitsstrafe bis zu einem Jahr oder Geldstrafe ein.

(2) Straflosigkeit tritt ein, wenn der Täter die falsche Angabe rechtzeitig berichtigt. Die Vorschriften des § 158

Abs. 2 und 3 gelten entsprechend.

Para. 161 StGB (German Criminal Code): False Statutory Declarations Due to Negligence

(1) If a person commits one of the offences listed in sections 154 through 156 negligently the penalty shall be imprisonment not exceeding one year or a fine.

(2) The offender shall be exempt from liability if he or she corrects their false testimony in time. The provisions of section 158 (2) and (3) shall apply accordingly.

Die vorstehende Belehrung habe ich zur Kenntnis genommen:

I have read and understood the above official notification:

Aachen, October 7, 2024
Ort, Datum/City, Date

Unterschrift/Signature

Abstract

Mastering academic writing is a fundamental challenge for many students, especially for non-native English speakers. Effective writing requires not only a strong comprehension of grammar and vocabulary but also the ability to structure ideas coherently and maintain a formal tone. Students often encounter difficulties in mastering academic writing, which can interfere with their academic progress and development. This thesis presents the development and evaluation of WILLM, an academic writing improvement tool based on large language models (LLMs) to provide immediate and long-term writing support. WILLM offers features such as detailed feedback on various writing elements, such as grammar, vocabulary, organization, coherence, and writing style along with automated scoring, and active recall quizzes and personalized reviews designed to reinforce learning over long-term. A three-week user study with 19 non-native English-speaking participants was conducted to evaluate WILLM’s effectiveness. Participants engaged with the tool’s features and completed pre- and post-tests to measure improvements in their academic writing skills. Additionally, they filled out a system usability scale (SUS) used to measure WILLM’s usability and overall user experience. Results indicated that while users showed some practical improvements, particularly in reducing errors and increasing scores, these gains were not statistically significant across all writing elements. The long-term improvement features did not demonstrate a significant impact on writing proficiency. The evaluation of SUS suggests that users found WILLM highly usable despite these challenges. We argue that further refinement of the tool, particularly in long-term improvement features, is essential for enhancing its impact on academic writing development.

Contents

1	Introduction	1
1.1	Motivation	1
1.2	Thesis Goals	4
2	Background and Related Work	6
2.1	Background	6
2.1.1	Academic Writing	6
2.1.2	Writing Assessment	7
2.1.3	Academic Writing Improvement	8
2.1.4	Large Language Models	9
2.1.5	Semantic Text Representation and Vector Database Retrieval	11
2.1.6	Gamification	12
2.2	Related Work	13
2.2.1	Writing Assessment	13
2.2.2	Writing Improvement Tools	14
2.2.3	Long-Term Memory Enhancement	15
2.2.4	Gamification	17
3	Concept	19
3.1	Requirement Analysis	19
3.1.1	Functional Requirements	19
3.1.2	Non-Functional Requirements	21
3.2	System Design	22
3.2.1	Components	22
3.2.2	Immediate Writing Improvement	24
3.2.3	Long-Term Writing Improvement	25
3.2.4	Gamification	31
3.3	Conceptual Limitations	31
3.3.1	Scope and Target Audience	31
3.3.2	Limitations for LLMs	31
4	Realization	33
4.1	Architectural Overview	33
4.1.1	System Components	33
4.1.2	Model Selection	36
4.2	Prompt Realization	36
4.2.1	Immediate Writing Improvement Prompts	36

4.2.2	Long-Term Writing Improvement Prompts	37
4.3	User Interface	48
4.3.1	Immediate Writing Improvement Features	48
4.3.2	Long-Term Writing Improvement Features	49
4.3.3	Gamification	49
4.4	Decisions and Limitations	50
4.4.1	Model Choice for Users	51
4.4.2	Cost Optimization	51
4.4.3	Testing Constraints	51
5	Evaluation	53
5.1	Methodology	53
5.1.1	Study Design	53
5.1.2	Participants	54
5.2	Results	55
5.2.1	Immediate Writing Improvement	55
5.2.2	Long-Term Writing Improvement	62
5.2.3	Gamification	65
5.2.4	System Usability Scale	66
5.3	Discussion	66
5.4	Limitations	69
6	Conclusion and Future Work	70
6.1	Contributions	70
6.2	Improvements and Future Work	71
A	Writing Rubric	i
B	Gamification	v
C	System Usability Scale and Detailed Results	vii

List of Figures

3.1	The use-case diagram of our system, depicting user interactions and functional offerings. Solid lines describe use cases, dashed lines in direction of the use case describe optional extensions and dashed lines in the opposite direction of the use case describe inclusions.	20
3.2	The proposed system overview of WILLM. Arrows denote an established communication channel between the components [Fla].	23
3.3	The sequence diagram depicting the AI evaluation of our system. . . .	25
3.4	The sequence diagram depicting the question generation of our system.	26
3.5	The sequence diagram depicting the quiz generation of our system. . .	28
3.6	The diagram depicting the quiz design of our system.	29
3.7	The sequence diagram depicting the review generation of our system. .	30
4.1	The architecture overview of WILLM [Fla].	33
4.2	The system and user prompt for the correction of grammar and vocabulary issues with GPT-4o.	38
4.3	The system and user prompt for correcting grammar and vocabulary issues with GPT-3.5 on sentence level.	39
4.4	The system prompt and part 1 of the user prompt for the correction of organization, coherence, and writing style issues.	40
4.5	Part 2 of the user prompt for the correction of organization, coherence, and writing style issues.	41
4.6	The system and user prompt for the assessment of the five writing elements through scoring.	42
4.7	The system prompt for the generation of the question types.	42
4.8	The user prompt for the generation of the revision question.	43
4.9	The user prompt for the generation of the synonyms question.	43
4.10	The user prompt for the generation of the organization question. . . .	44
4.11	The user prompt for the generation of the coherence question.	45
4.12	The user prompt for the generation of the academic sentence question.	45
4.13	The user prompt for the generation of the argument strengthening question.	46
4.14	The system and user prompt for the question selection based on spaced repetition principles.	46
4.15	The system prompt for the generation of organization or coherence tips.	47
4.16	The user prompt for the generation of organization tips.	47
4.17	The user prompt for the generation of coherence tips.	47

4.18	An example of how the text evaluation could look like after clicking on the button "AI Evaluation".	48
4.19	(left) An example of how the review could look like after clicking on the button "Review". (right) An example of a user profile including the score comparison on the left, and the gamification progress on the right.	49
4.20	An example of how a question could look like inside the quiz.	50
4.21	An example of how the user profile looks like.	50
5.1	Average number of identified issues per writing element by score. . . .	56
5.2	Score distribution for each writing element. The bar in the center of each violin represents the interquartile range, which includes 50% of the scores and the small line shows the median score.	57
5.3	Pre-test vs post-test grammar and vocabulary mistakes.	59

1 Introduction

1.1 Motivation

Academic writing presents a significant challenge for students, particularly due to its complex nature, which demands advanced writing skills. This difficulty is universal, affecting both native and non-native English speakers [BL20]. Research highlights that students often struggle with language use, coherence, and cohesion, making errors in grammar, vocabulary, and sentence structure, largely due to a lack of awareness and unintentional mistakes [Jab18] [Sun09] [Fit18] [AB15] [AT18]. In addition to this, academic writing also requires clarity, precision, formality, and cohesion, with an emphasis on concision and factual accuracy [CGA19]. For non-native speakers, these difficulties are compounded by the need to master not only the technical aspects of writing but also the rhetorical and cultural conventions of academic English, which can vary significantly from their native languages [AF12]. Furthermore, discrepancies between the writing mechanisms of their first and target languages often result in a low-quality writing product [CGA19].

However, students often receive inconsistent or unclear feedback from instructors, leaving them confused about how to improve [BL20] [AB15]. Many instructors assume that students have already developed strong academic writing skills in previous education, but this presumption often leaves students feeling overwhelmed and uncertain about meeting academic expectations [RMA23]. Beyond grammar and structure, students also struggle with academic conventions such as organizing their writing, rephrasing content, and incorporating external sources while maintaining their own academic voice, which are essential for success in higher education [LM21] [Akb20]. These barriers create anxiety and undermine students' confidence, making it even more difficult to produce academic work [RMA23] [CGA19]. While these challenges in mastering academic writing are evident, they are further influenced by the way writing is assessed in academic settings.

Writing assessment faces several significant challenges, especially for non-native English speakers. One major issue is feedback, as students often receive insufficient or delayed responses from instructors, making it difficult for them to improve their writing. Time constraints intensify these problems, as non-native speakers spend more time revising their work [GJPK22][ZMSR19]. The absence of standardized rubrics for academic writing assessment, particularly for complex tasks like thesis proposals, also adds to the confusion. While rubrics for essays exist, they are not always adequate for assessing higher-level academic work, leaving students unclear about expectations [GJPK22][ZMSR19]. This inconsistency can lead to unfair evaluations for

1 Introduction

students [GJPK22]. To address these challenges, writing improvement tools can offer personalized support, helping students enhance their academic writing and easing the assessment burden on educators. These tools play an important role in resolving the issues in both academic writing and its assessment but they too have various issues.

While numerous writing tools aim to assist students with improving their writing, each comes with limitations that reduce their effectiveness in academic contexts. Wordtune, for example, lacks the specialized academic vocabulary necessary for scholarly writing, and its overwhelming number of rewrite options often makes it difficult for users to select the most appropriate revision. Additionally, there is concern about users becoming overly dependent on artificial intelligence (AI) generated suggestions, which could further hinder the development of their writing skills [ZXC24]. Grammarly [Gra], while popular, primarily addresses surface-level issues like grammar, spelling, and punctuation, neglecting deeper aspects such as content, meaning, rhetorical awareness, coherence, argumentation, and organization [Ish23] [CCBC22]. It also struggles with more complex sentence structures, often failing to correct run-on sentences or identifying unclear sentences as correct. This is largely because Grammarly’s pattern-based system cannot account for the meaning behind the structures it evaluates [BMK24]. Moreover, its suggestions for wordiness are sometimes too general, failing to consider the specific context or stylistic choices of the writer [BMK24]. Students report confusion about why errors were flagged and struggle to understand how to respond to the tool’s feedback [GR18]. ProWritingAid [Pro] faces similar shortcomings, particularly when it comes to addressing persistent language issues or detecting errors in poorly constructed sentences, sometimes offering misleading or incomplete feedback [SBA] [AMT19]. Another concern is the potential over-reliance on AI tools in academic settings, as they may impact the validity of test results, like the Test of English as a Foreign Language Internet Based Test (TOEFL iBT) [ETS], the International English Language Testing System (IELTS) [IEL], and the Pearson Test of English (PTE) Academic [PTE], when students use AI to generate or refine their answers, raising ethical concerns about academic integrity [HLS⁺23] [N⁺23] [TG24]. Furthermore, most of these tools lack long-term enhancement techniques, focusing primarily on short-term fixes rather than promoting durable improvement in writing skills. Ongoing research and development are necessary to address these emerging challenges and ensure the ethical application of AI in academic writing [KA24]. In addition to existing writing improvement tools, gamification has also been explored.

Typically, gamification aims to address the challenge of how to increase student engagement, motivation and academic achievement. However, in our thesis it is only examined as a means to measure student engagement. Nonetheless, the broader implications of gamification, such as its impact on non-native English speakers, remain relevant. For instance, the ability to write in academic English may be influenced by factors such as engagement, motivation, and other emotional elements [AF12]. Furthermore, the optimal type and quantity of gaming components to be integrated into educational gamification remain subjects of ongoing debate and there is still a lack of

consensus among experts in the field [FR15]. Also, some researchers have pointed out potential drawbacks of gamification, such as the risk of oversimplifying complex subjects, the perception of learning activities as mere games, the insufficiency of games to boost performance on their own, and the inability of games to address learning challenges solely [AZA13]. For instance, Faiella et al. [FR15] reviewed literatures and concluded that the gamification effects are greatly dependent on the users using it and Hanus et al. [HF15] found that participants in the gamification program exhibited lower levels of motivation, empowerment, and satisfaction than participants in a non-gamified community. A further challenge lies in the necessity for designers to conceive novel ideas tailored to each educational subject, diverse student needs, and the judicious selection of gamification components [SNO22] to ensure the efficacy of gamified experiences.

1.2 Thesis Goals

To address the problems outlined in the problem statement above, this thesis integrates large language models (LLM) into a system designed to improve academic writing. The main contribution of this thesis, WILLM: A System for Academic Writing Improvement based on Large Language Models, is to investigate how effectively LLMs can enhance students' writing skills, not only in the short-term but also through long-term learning techniques. This leads to the following core research questions:

- **Research Question 1:** Does continuous written feedback on grammar, vocabulary, organization, coherence, and writing style along with automated scoring from LLMs improve academic writing over time?
- **Research Question 2:** How effective are active recall quizzes from LLMs in improving long-term writing?
- **Research Question 3:** How effective are personalized reviews from LLMs in improving long-term writing?

To achieve these goals, the system meets several concrete requirements:

- **Immediate Feedback:** Continuous, automated feedback on grammar, vocabulary, organization, coherence, and writing style.
- **Long-Term Enhancement:** Active recall quizzes combined with spaced repetition and personalized reviews.

Through these contributions, this thesis aims to demonstrate the effectiveness of using LLMs for academic writing improvement while addressing both immediate writing issues and long-term skill development.

We now provide a brief overview of the structure of this thesis and its upcoming chapters. In Chapter 2, we begin by explaining the background terminology, including academic writing, assessment methods, along with the techniques applied in this thesis and an overview of gamification. This is followed by a review of related work. Chapter 3 presents the concept of WILLM, including a concrete requirement analysis, system design and conceptual limitations. In Chapter 4, we describe the realization and implementation of our system whereas its evaluation is discussed in Chapter 5. Finally, we conclude our thesis in Chapter 6 with a summary of our work and future research considerations.

2 Background and Related Work

2.1 Background

This chapter provides a comprehensive foundation for understanding the various components of academic writing improvement using LLMs. We begin by defining academic writing, outlining its significance in education and research, and exploring its main characteristics. Following this, we delve into the methods and challenges of writing assessment, discussing both formative and summative approaches, and the role of scoring. We then explore academic writing improvement strategies, focusing on immediate and long-term writing improvement approaches such as correction, scoring, quizzes and reviews. Subsequently, we introduce LLMs, outlining their capabilities, relevance to academic writing, and the technical aspects of fine-tuning and prompting. Finally, we discuss the role of gamification in enhancing user engagement and motivation through elements like experience points (XP), levels, badges, and achievements.

2.1.1 Academic Writing

Academic writing is a formal and structured style of communication used by researchers, educators, and students to convey ideas, information, and research within the academic community. It is characterized by its focus on evidence-based arguments, logical reasoning, and objective analysis. Academic writing can be categorized into two main types: student academic writing, often used as a form of assessment in educational settings and expert academic writing, intended for publication in academic journals or books. Both types adhere to standards of logic, clarity, formality and precision, which can present challenges for students to master [IE] [oS] [Smi]. A logical and clear structure is essential, typically following an introduction, body and conclusion to ensure coherence and help the reader follow the argument or thesis statement. Another important standard in academic writing is formality, requiring formal vocabulary, complex sentences and an impersonal, objective tone that focuses on facts and evidence rather than personal opinions. Writers must support their arguments with credible sources, properly cited and engage in critical thinking to analyze and synthesize information into well-supported conclusions [Smi] [oS]. In summary, academic writing is a formal, objective, and structured form of communication essential in education and research. Its characteristics set academic writing apart from other forms of writing and make it a critical skill for students and professionals in academic fields. Understanding these elements of academic writing provides the foundation for another crucial aspect in writing: writing assessment.

2.1.2 Writing Assessment

Writing assessment refers to a field of study that delves into the theories and practices guiding the evaluation of a writer's performance or potential through writing tasks. It employs techniques and methods to assess student academic writing and learning outcomes. Within this context, various elements such as grammar, language use, structure, and writing style serve as standards for evaluating the quality and effectiveness of writing.

Methods

Corresponding writing assessment methods include assessing writing-related skills such as grammar and vocabulary through non-writing tasks and evaluating actual written work for its grammar, language use, structure, style, etc. In our investigation, the former approach is not applicable; hence we focus exclusively on the latter. This approach can be separated into two distinct methods, each serving a different purpose. The difference in writing assessments can be seen in the two approaches: formative and summative evaluation. Relying on the distinctions highlighted by [Stu01], formative evaluation is a continuous, proactive approach designed to support revisions and improvements throughout the instructional process. This type of assessment is focused on enhancing learning and teaching in real-time, often through activities like drafts, peer reviews, and self-assessments. On the other hand, summative evaluation takes a more static, retrospective approach, assessing the effectiveness of instruction after it has been completed. This method typically aims to provide a final evaluation for an external audience, often resulting in grades or scores through final essays, standardized tests, or other academic work. In accordance with these definitions, this thesis aligns itself with both assessment methods by introducing a system that evaluates and improves textual inputs from users, offering constructive feedback in a formative manner while also capable of providing a definitive evaluation in the form of a summative score.

Scoring

Scoring in this context refers to the evaluation of writing quality through the application of as previously explained predefined standards to measure various aspects of writing. Consequently, a writing rubric often serves as a rating scale that shows the range of possible scores and the criteria required to achieve each one. Rubrics are scoring tools that list specific components and expectations for an assignment, making them applicable to a variety of tasks such as tests, research papers, group projects, portfolios, and presentations. Well-known examples of writing rubrics include those used in standardized tests such as TOEFL iBT [ETS], IELTS [IEL], and PTE Academic [PTE]. The purpose of these standardized tests is to evaluate the test takers proficiency in the four fundamental language skills of reading, writing, speaking, and listening in English. All of the exams are appropriate for non-native English speakers who want to demonstrate their ability in academic or professional settings and are

2 Background and Related Work

widely accepted for university admissions.

Additionally, various scoring methodologies exist within the realm of writing assessment, with the choice of a particular method depending on the goal and specific context. Given our focus on a tool designed for the improvement of academic writing, the discussion herein explores various scoring approaches relevant to this aim.

- **Analytic Scoring:** Analytic scoring breaks down the writing into several components such as grammar, language use and style where each is scored individually. The scores can be combined for a total score or reported separately. This method provides more detailed feedback on specific areas of strength and weakness.
- **Holistic Scoring:** Unlike analytical scoring, holistic scoring involves assigning a single, overall score to a piece of writing. The scorer evaluates the writing as a whole while considering its general quality and assesses how effectively the various elements of the writing work together to convey meaning. Holistic scoring is often used in contexts where time is a constraint.

In addition to these traditional methods, automatic essay scoring (AES) has emerged as a modern approach that leverages machine learning algorithms to evaluate writing. Tools like Grammarly utilize AES to assess grammar, style, and other aspects of writing, providing immediate feedback. AES can serve both analytic and holistic purposes, offering detailed feedback on specific elements or an overall assessment.

In summary, scoring in academic writing involves various writing rubrics and methodologies, such as analytic, holistic scoring, and AES. Having established the importance of effective writing assessment, the next step is to explore strategies for improving academic writing skills, both immediately and over long-term.

2.1.3 Academic Writing Improvement

Academic writing improvement refers to a broad range of activities aimed not only at correcting surface-level errors such as grammar and spelling mistakes but also at enhancing the text's overall composition to match academic style. It focuses on both enhancing the current text and the improvement from one writing task to a subsequent one. With this understanding, academic writing improvement can be categorized into two areas: immediate and long-term improvement.

Immediate Writing Improvement

Immediate writing improvement involves quickly correcting mistakes and enhancing writing without necessarily contributing to long-term writing skill development. Corrections in grammar, vocabulary, organization, coherence and writing style are important as they ensure alignment with academic standards. Scoring, which provides quick insights into the strengths and weaknesses of the text, is also a part of immediate writing improvement.

Long-Term Writing Improvement

Long-term writing improvement refers to techniques, such as active recall, spaced repetition, and reflection, aimed at enhancing writing over a long-term perspective.

Active recall is a learning technique that involves retrieving information from memory rather than simply looking at it and has been shown to significantly enhance retention [DRM⁺13]. On the other hand, spaced repetition refers to the practice of distributing learning over time which also has been found to benefit long-term retention more effectively than massed learning sessions [DRM⁺13]. By increasing the intervals between recall sessions, spaced repetition combats the forgetting curve, ensuring that information remains accessible over longer periods. In the context of academic writing improvement, these principles can be implemented through quizzes that are designed to reinforce learning by testing the user’s knowledge of academic writing principles, thus promoting retention and retrieval of information. By regularly engaging in quizzes, users can strengthen their understanding of academic writing and improve their ability to apply it over time.

Additionally, reflection involves thinking critically about one’s own ideas and it is essential for deep learning and continuous improvement [Cha19]. In the context of writing, reflection can be implemented through reviews. Reviews allow writers to revisit past work, critically assess their mistakes, receive constructive feedback, and track their progress over time. This process not only helps in identifying recurring issues but also supports the development of improved writing skills.

Tools for Writing Improvement

Since using the approaches mentioned above independently — without relying on machines — requires a great deal of self-effort, repetition, and feedback from an outsider, the process can become time-consuming and challenging. To address these difficulties, writing improvement tools have been developed with a two-fold goal: to elevate the quality of writing and to simultaneously improve the writing skill of the user. Many of these tools leverage the recent advancements in AI and natural language processing (NLP) technologies to implement features such as grammar and spell checking, paraphrasing, enhancing structure and coherence, preventing plagiarism, providing tailored feedback with suggestions for improvement, and relying on scoring and gamification. Tools like Grammarly, ProWritingAid and Quillbot [Quia] exemplify this by analyzing the tone and style of writing and offering nuanced corrections and suggestions tailored to the context, such as academic writing. Since NLP techniques are becoming more popular these days, research also focuses on leveraging LLMs - a subset of NLP - for this task. However, despite the potential of LLMs, there are only a few writing improvement tools using them as discussed in Section 2.2.2, related work.

2.1.4 Large Language Models

LLMs are advanced AI systems designed to understand, generate, and manipulate human language. These models are trained on vast amounts of text data, allowing

2 Background and Related Work

them to grasp the context and complexities of language. LLMs leverage deep learning techniques to predict the next word in a sentence, given all the previous words, thus enabling them to generate coherent and contextually relevant text. The development and applications of LLMs have gained considerable attention in research. One of the foundational breakthroughs here was introduced by Vaswani et al. [VSP⁺17] which presented the Transformer architecture which revolutionized the way LLMs are developed by introducing the concept of attention mechanisms that allow models to weigh the importance of different words in a sentence, enabling more subtle understanding and generation of text. Such developments have opened the way for models such as GPT-3.5, GPT-4 or GPT-4o and applications like ChatGPT [Ope22], which has gathered global interest [FHNZC⁺23] by allowing user interaction through prompts. It is based on a decoder-only design, meaning it is trained on the next token prediction tasks from left to right generation. Other designs include encoder-only models like BERT [DCLT18] or encoder-decoder models like T5 [RSR⁺20]. LLMs have achieved remarkable milestones, including surpassing human abilities in tasks like summarization [PGW23], excelling in various exams [AAA⁺23], and demonstrating advanced reasoning through techniques like prompt engineering and Chain-of-Thought [WWS⁺22]. Building on the success of LLMs, the next step involves adapting these models for specific tasks to improve their performance such as with fine-tuning and prompt engineering.

Fine-tuning a LLM involves adjusting a pre-trained model's parameters to adjust it for a specific task or dataset, enhancing its performance on that particular application. Typically for this, a model that has learned general language patterns from a broad dataset is used and is then further trained on a smaller, task-specific dataset to adapt its knowledge. Through fine-tuning, LLMs are refined to better produce text in line with the intended task. However, fine-tuning a LLM for downstream task adaptation requires hundreds or thousands of examples [DCLT18]. Additionally, this technique may be challenging in an educational setting since there is limited generalization between learning objectives or students' needs and training data could be hard to obtain [KYT⁺23]. Furthermore, Radford et al. [RWC⁺19] showed that there is no need to condition the output of the language model on the target task, instead simple human-like instructions can be given and Kwon et al. [Kwo23] mentions that the focus has shifted from designing task-specific architectures or fine-tuning model weights to prompt engineering since the recent rise of LLMs.

Prompt Engineering

In the context of LLMs, prompt engineering refers to the method of constructing prompts which provide these models with an initial input to guide their output generation. Prompting can be divided into zero-shot, one-shot and few-shot prompting, each of which involves a varying number of provided examples of the intended task. Zero-shot prompting involves providing the model with a task or question without any additional examples, relying on its pre-trained knowledge. This can be advantageous when needing a response that includes a broad understanding of a topic. Also, accord-

ing to [TJ23], a lot of research has shown that LLMs e.g. understand user generated content in a zero-shot manner. On the other hand, one-shot prompting includes a single example to illustrate the desired task whereas few-shot prompting extends this concept by offering several examples to the LLM, refining the model’s understanding of the task. This is particularly relevant when e.g. improving the accuracy of its responses and needing a specific output for a specific task. Each prompting technique is based on the task and the availability of relevant examples.

In the context of LLMs, prompt engineering has emerged as a critical technique, enabling these models to generate more precise and contextually appropriate responses. However, another crucial aspect of language processing focuses on how textual information can be efficiently and meaningfully represented in a machine-readable format. This brings us to the concept of semantic text representation and retrieval techniques.

2.1.5 Semantic Text Representation and Vector Database Retrieval

This section is about the foundational technologies that enable machines to understand and work with human language in a structured and efficient manner. We will explore text embedding models, which are needed for converting text into dense vector representations that encapsulate semantic meaning. These embeddings are numerical representations of text, where words, sentences, or entire documents are transformed into high-dimensional vectors. The key advantage of this transformation is that it allows similar pieces of text to be positioned closer together in the vector space, thereby capturing the subtle relationships between different texts [Rei19] [LZZ⁺23].

These embeddings serve as the basis for similarity search, a technique that allows us to find text data that is most similar to a given query. In a similarity search, the query text is first converted into its corresponding embedding. The search then involves comparing this embedding to a large set of embeddings to identify those that are closest in the vector space, effectively retrieving the most relevant or similar pieces of text. This method is widely used in applications like document retrieval and recommendation systems.

To efficiently store and retrieve these embeddings during a similarity search, vector databases come into play. Vector databases are specialized systems optimized for managing and querying high-dimensional vectors. Unlike traditional databases that handle structured data with fixed schema, vector databases are designed to handle the complexities of similarity search in a large scale to enable fast and accurate retrieval of relevant data based on vector distances. They support algorithms like approximate nearest neighbor search, which speeds up the process by finding near matches rather than exact ones, and can integrate with applications that rely on real-time text retrieval [Nex] [Tri24] [PWL24] [Mar24].

After exploring the capabilities of LLMs, text embedding models, retrieval techniques and their potential to enhance academic writing, we are going to discuss another component of learning enhancement tools: gamification.

2.1.6 Gamification

Gamification refers to the use of game elements, methods, and frameworks in non-game contexts to engage and motivate users. Over recent years, the application of gamification has grown significantly through its increasing popularity and the positive outcomes it has achieved, particularly in education [Det11] [TdCdS⁺19]. By systematically integrating game-like features into various activities, gamification aims to enhance user experience and effectiveness in achieving specific goals. Specifically, in an academic writing improvement context, it aims to increase motivation, interaction, and thus performance among learners. By incorporating game-like features regarding this context such as points, badges, levels, rewards and feedback, as highlighted by [SNO22], developers can create a more engaging and interactive learning environment. This can lead to increased engagement and motivation among users, as they are more likely to participate actively in their learning when presented with more fun and rewarding tasks. While gamification can enhance engagement and motivation, it's important to carefully balance external rewards with intrinsic motivation [HF15]. The introduction of external rewards or game elements could potentially shift the focus from the enjoyment of the activity itself to earning rewards. Thus, in order to ensure the effectiveness of gamification, it is important to define clear objectives, understand the audience, choose the right game mechanics, offer rewards, and encourage continuous learning.

In summary, this section provided a comprehensive overview of the key components essential for academic writing improvement using LLMs, from understanding the fundamentals of academic writing and assessment to exploring the potential of modern AI tools and gamification in learning environments.

2.2 Related Work

This section provides an overview of the research that supports the development of WILLM. We review literature on writing assessment, focusing on formative and summative approaches as well as automated scoring systems. We then examine existing writing improvement tools, discussing their capabilities in providing real-time feedback. Additionally, we explore techniques for long-term memory improvement, such as spaced repetition, quizzes, and reflective learning, which are important for durable writing enhancement. Finally, we investigate the role of gamification in fostering user engagement and motivation in educational tools.

2.2.1 Writing Assessment

Writing assessment has been explored for a long time, finding its parallel in the contrasting methodologies of formative and summative assessment [Stu01]. More recently, Mohamadi [Moh18] implies that using engaging technology along with appropriate formative and summative assessment strategies is a powerful way of making learning efficient as shown by their participant’s writing from pre- to post-tests. With regard to the improvement of students academic English writing, Chen et al. [CZ17] suggest that formative assessment may be more effective than summative assessment and Horstmannshof et al. [HB13] concluded that learners were in favor of timely and focused feedback provision through formative assessment to help them acquire the necessary skills.

One way to provide formative or summative writing assessment is through scoring. As explained in section 2, there are several methods for this task. Analytic and holistic scoring were evaluated by human scorers in the work by Hunter et al. [HJR96] and Ghalib et al. [GAH15] revealing that while both approaches yield consistent results, analytical scoring offers more accurate and consistent scores.

AES systems have further revolutionized the field by utilizing advanced machine learning techniques, particularly LLMs, to assess writing with remarkable accuracy. For instance, as highlighted by Xie et al. [XCK⁺22], neural networks have set a new standard in AES, surpassing traditional models. Following this achievement, other researchers explored fine-tuning LLMs like BERT [DCLT18] for writing assessment tasks which yielded remarkable results [WWLL22] [YCW⁺20]. Other work include [DZY17] and [WHN⁺22] where they predict an overall holistic score with pre-trained language models that closely matches human judgment.

That said, recent work has focused on the capabilities of LLMs for both formative and summative assessment tasks. These models leverage techniques like prompt engineering, where specific inputs instruct the models to generate or evaluate text based on predefined rubrics. For example, these models offer capabilities like few-shot learning [BMR⁺20], meta-prompting [YCRI22], and enhanced prompt engineering strategies that improve model performance across tasks. Well-known writing aids like Grammarly and tools like ChatGPT use LLMs to automatically assess grammar, coherence, and structural quality, providing real-time feedback that mimics human judgment

2 Background and Related Work

[AA23]. Mizumoto et al. [ME23] and Mansour et al. [MAEE24] were among the first to employ GPT-based models for writing assessment, confirming that LLMs can predict overall writing quality in ways that closely align with human evaluators. To support assessments within WILLM, a custom writing rubric was developed, blending aspects of the IELTS [IEL], PTE Academic [PTE], and TOEFL iBT [ETS] rubrics. This approach is supported by various studies that have employed similar rubrics to evaluate the efficacy of AI in writing assessment. For instance, Tran et al. [Tra23] applied TOEFL band descriptors in assessing essays, while Song et al. [SS23] assessed student work using IELTS band descriptors in an evaluation of AI-assisted language learning tools. Further studies, such as those by Geçkin et al. [GKC23], explored the potential of generative AI, like ChatGPT, for replacing human raters, using a custom holistic rubric tailored for second-language writing assessment. Other research has shown the effectiveness of rubrics adapted from standardized tests such as TOEFL for guiding the development of learners' writing skills in academic and test-preparation settings [MH21]. Lastly, Etaat [Eta24] designed a rubric based on a task from IELTS to examine how AI-based applications affect specific aspects of writing, such as task fulfillment, organization, and grammar.

While writing assessments provide valuable insights into a learner's current skill level, AI-powered tools support writing improvement by offering real-time feedback.

2.2.2 Writing Improvement Tools

Writing improvement tools provide assistance in the creation, editing and structuring of written content and also enhancing it with clarity and consistency. They serve for a wide range of purposes, from creative storytelling to academic writing, increasing productivity and enhancing user writing skills over time.

Widely-used tools such as Microsoft Word [Mic], Google Docs [Goo], and DeepL [Dee] have long been relied upon for basic grammar and spell-checking functionalities, as well as formatting and document collaboration features. Recently, these platforms have integrated AI-based features to further enhance their utility. For example, Microsoft Word and Google Docs now offer AI-driven suggestions for style, tone, and sentence structure, while DeepL uses advanced AI for translation and language improvement, providing users with more accurate translations that preserve the context.

Recent developments in AI have led to the creation of sophisticated writing tools to improve various aspects of academic writing. Grammarly is perhaps one of the most widely used tools, offering comprehensive grammar checks, plagiarism detection, and writing assessments [GRB19]. It also provides AI-driven suggestions for paraphrasing and content generation [Fit21]. Research shows that Grammarly significantly improves students' cohesion, coherence, lexical resources, and grammatical accuracy, making it a valuable tool in academic settings [Tra23] [DH24]. Similarly, Ishak [Ish23] found that Grammarly plays a significant role in reducing grammatical errors and improving writing quality for Algerian students.

Quillbot is another powerful tool, known for its paraphrasing, summarizing, and rephrasing capabilities. Studies by Kurniati et al. [KF22] and Ariyanti et al. [A⁺21] demonstrated its effectiveness in aiding language development and improving student writing scores, particularly through its user-friendly interface and writing features. Like Grammarly, Quillbot has been found to significantly improve cohesion, coherence, lexical resources, and grammatical accuracy. The work by Dja'far et al. [DH24] further confirmed the effectiveness of Quillbot when integrated into academic settings, where it significantly enhanced students' writing proficiency.

ChatGPT has also proven to be a valuable resource in academic writing improvement. Song et al. [SS23] found that ChatGPT offers real-time feedback on grammar, vocabulary, sentence structure, and coherence. Its ability to track participants' progress contributed to significant improvements in both writing skills and motivation. Similar to Grammarly and Quillbot, ChatGPT has been shown to enhance students' grammar, coherence, and overall writing proficiency in academic contexts.

Another promising tool, Automated Feedback, introduced by Huang et al. [HtW], provides instant formative feedback to help students develop academic writing skills. While its effectiveness is still under study, the paper suggests it could be useful for better academic writing outcomes.

WordTune [Wor] and InstaText [Ins] have shown significant positive effects on task fulfillment, organization, grammar, and lexis. Non-native English speakers receiving automatic corrective feedback from these tools experienced notable improvements in their overall writing skills [Eta24]. Additionally, Zhao et al. [ZXC24] highlighted the usefulness of WordTune in helping Chinese international students rewrite text in formal language, although some concerns were raised about dependency on AI tools and their limitations in handling specialized academic vocabulary.

A study by Han et al. [HZN21] explored the impact of technology-enhanced writing environments through a comparative study of different models. Their findings indicate that automatic writing evaluation platforms like Pigai [PIG], which provide feedback on lexical, syntactic, semantic, and content-based parameters, significantly improve writing performance and support students' self-regulated learning strategies.

While AI-powered writing improvement tools have shown remarkable progress in enhancing immediate writing skills through real-time feedback, paraphrasing, and grammar corrections, there is relatively little research focused on improving academic writing from a long-term perspective.

2.2.3 Long-Term Memory Enhancement

Long-term memory is crucial for academic writing improvement, as it allows students to retain and apply writing principles over time. While AI-powered tools focus primarily on immediate corrections, there has been little research into integrating long-term memory enhancement techniques into academic writing systems. Methods like spaced repetition, quizzes, and reflective learning offer the potential to improve writing proficiency in the long term.

2 Background and Related Work

Spaced repetition is one of the most effective methods for enhancing long-term memory. It adapts to individual learning curves by scheduling reviews at increasing intervals, ensuring that information is revisited just before it is likely to be forgotten. This technique, widely used in tools like Duolingo [Duo], Anki [Ank] and Quizlet [Quib], helps students memorize facts and concepts efficiently. As explored in the work by Munday [Mun17] and Thilakarathna et al. [TG24], AI-powered spaced repetition algorithms in these tools significantly improve memory retention. Personalized quizzes also allow students to better master complex skills over time, which could be adapted to reinforce academic writing principles.

In addition to spaced repetition, regular quizzes are another proven method for enhancing long-term retention. Studies by Dunlosky et al. [DRM⁺13] and Refnita [Ref17] highlight how frequent quizzes help reinforce learning by promoting active recall and retention. This method is particularly beneficial for complex skills like academic writing, where continuous assessment helps students retain and apply writing strategies more effectively. Building on this concept, Ruan et al. [RJX⁺19] introduces a dynamic AI-based system that improves long-term retention through personalized quizzes. QuizBot leverages forgetting curves and adapts its question selection to the student's ability, leading to better recall rates compared to traditional methods. By focusing on factual knowledge in areas such as science and vocabulary, QuizBot demonstrates how adaptive systems can enhance long-term learning outcomes, which can be similarly applied to writing instruction. Furthermore, Liu et al. [LCR12] proposed an automatic question generation system to enhance writing by prompting users with questions designed to stimulate critical thinking and refinement of their content. This system encourages deeper engagement with the material, fostering long-term retention of writing principles through active interaction.

Another important strategy for reinforcing memory and enhancing long-term skill development is reflective learning. According to Chang [Cha19], reflection allows students to identify gaps in their understanding and revisit concepts, which helps solidify what they have learned. The study advocates for integrating reflection-based assessments, where students reflect on their past performance and areas for improvement. This approach can be especially effective for academic writing, where reflection on previous work can guide students toward developing better writing habits over time. Additionally, another study about reflective learning identified nine key reflective strategies including reviewing notes, memorizing useful sentences, and reflecting on mistakes from quizzes [Nan23]. These strategies are particularly relevant to language learning and writing development, helping learners engage critically with their progress. Reflection based on feedback, especially from AI-driven systems, can significantly enhance long-term retention by fostering self-awareness and targeted improvement. AI-powered tools like ChatGPT also integrate reflective learning to help users track their progress and improve their writing skills. As demonstrated by Song et al. [SS23], ChatGPT not only provides real-time feedback on grammar, coherence, and vocabulary but also encourages users to reflect on their strengths and weaknesses through portfolio writing. This reflective practice enables learners to assess their progress continuously, leading

to long-term retention of writing skills and better self-regulated learning.

In summary, techniques such as spaced repetition, active recall, quizzes, and reflective learning have proven effective in enhancing long-term memory in various educational contexts. Integrating these methods into academic writing improvement systems could provide a solution for students to not only make immediate corrections but also retain and apply writing skills more effectively over time. While long-term memory enhancement techniques focus on strengthening the retention and application of writing skills, sustained engagement is equally important.

2.2.4 Gamification

Gamification offers a unique way to increase user motivation and interaction with writing tools, helping learners stay committed to their learning process. Specifically, Krause et al. [KMPW15] and Dichev et al. [DD17] declared that the main aims of gamification are to set goals that give learning a reason, improve specific skills, involve students and maximize learning. Kyewski et al. [KK18] explained that the most significant factor in learning is motivation. In order to achieve the desired outcomes mentioned above, there are several studies that showed that gamification has a positive impact. One of the most popular language learning application incorporating gamification is Duolingo. This also gets supported in the broad research area by Dehganazadeh et al. [DD20] explaining that Duolingo is the most investigated platform and very effective as outlined by Loewen et al. [LCI⁺19] by incorporating elements such as challenging tasks, reward incentives, systematic levels, and the ranking of users according to their achievements [STK⁺23]. A few examples that highlight the importance and usefulness of gamification are research done by Pechenkina et al. [PLO⁺17] who performed a study to analyze whether the use of a gamified mobile learning app influenced students' academic performance and boosted their engagement and Khaleel et al. [KAW20] who tried to increase student engagements in learning by conducting a gamification technique in programming language courses.

However, opinions on gamification as a learning enhancement strategies are two-folded. Faiella et al. [FR15] reviewed literatures and concluded that the gamification effects are greatly dependent on the users using it. Students who received instruction using conventional methods received the same score as those who completed the gamified exercises. Hanus et al. [HF15] found that participants in the gamification program exhibited lower levels of motivation, empowerment, and satisfaction than participants in a non-gamified community. Consequently, this thesis only aims to measure user engagement through gamification.

In this chapter, we provided a literature review about the current state of academic writing assessment, writing improvement tools, long-term memory enhancement, and gamification. We explored the methodologies of assessment and the importance of real-time feedback in improving writing quality. Additionally, we delved into various writing improvement tools, highlighting their effectiveness in short-term writing

2 Background and Related Work

enhancement, and discussed the potential for long-term memory enhancement techniques such as spaced repetition, quizzes, and reflective learning. Finally, we examined the role of gamification regarding user engagement and motivation, discussing both its benefits and challenges. This chapter sets the context for the concept of WILLM designed to leverage LLMs and integrate these techniques for both immediate and long-term academic writing improvement.

3 Concept

In this chapter, we will introduce the requirement analysis along with WILLM’s system design and its constituent building blocks. Section 3.1 covers the requirements our system needs to attain to achieve the goal of this thesis whereas Section 3.2 explains the design and architecture. Finally, Section 3.3 focuses on the conceptual limitations of WILLM.

3.1 Requirement Analysis

For building or modifying a system, it is essential to identify and define the requirements and specifications a system must attain based on user expectations and research insights. This process is covered by the requirement analysis. After thoroughly investigating and understanding these needs through related work in Chapter 2, we are going to dive deeper into the requirements for our system. This section is divided into two parts: Functional requirements and non-functional requirements.

3.1.1 Functional Requirements

Functional requirements typically describe through functions what the system should be able to perform. This is also depicted in the use-case diagram in Figure 3.1 which provides a visual representation of the functions which the user can interact with in our academic writing improvement system. We see the following functional requirements as essential for our system:

Immediate Writing Improvement Features

The immediate writing improvement features focus on the AI evaluation which is divided into two key components: correction and scoring.

- **Correction:** The correction feature, represented by the *Get Correction* component in Fig. 3.1, analyzes the user’s writing and provides corrections on various writing elements. Based on thorough research, these elements are grammar, vocabulary, organization, coherence and writing style. It includes the *Explanation* feature, which offers clear and accessible explanations for each correction, allowing users to understand the reason behind each suggested change. Users can optionally choose the language in which they receive these explanations, accommodating a diverse user base.

Requirement: This feature must deliver accurate and detailed corrections, categorize the type of mistake appropriately and provide thorough explanations

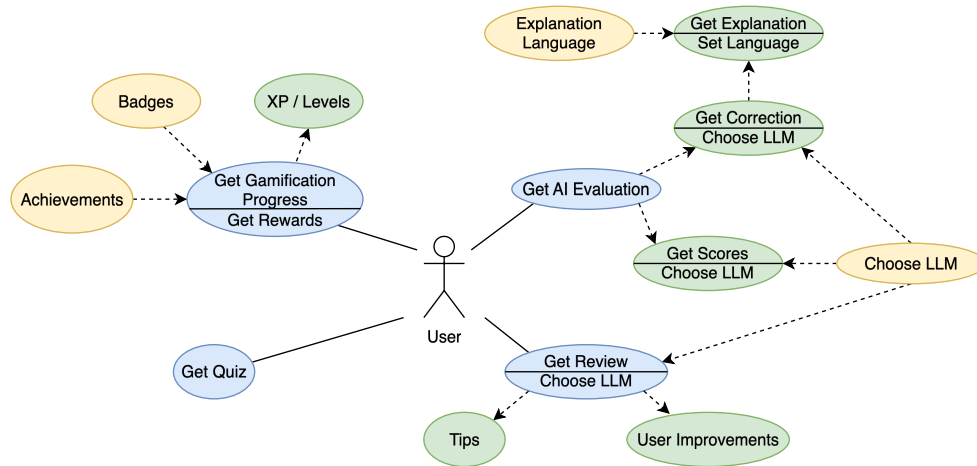


Figure 3.1: The use-case diagram of our system, depicting user interactions and functional offerings. Solid lines describe use cases, dashed lines in direction of the use case describe optional extensions and dashed lines in the opposite direction of the use case describe inclusions.

alongside the corrected version. Support for multiple languages is essential to ensure clarity and accessibility.

- Scoring:** The scoring feature, corresponding to the *Get Scores* component, is responsible for evaluating the user's writing against established academic standards. This feature must use solid writing rubrics as the baseline for assessment to ensure consistent and accurate evaluations.

Requirement: The scoring feature must provide evaluations that are accurate and aligned with academic standards, ensuring that users receive reliable feedback on their writing quality.

Both the correction and scoring features include the *Choose LLM* feature, which allows the user to select the LLM for their evaluation (see Subsection 4.1.2).

Long-Term Writing Improvement Features

- **Quiz:** The quiz feature, equivalent to the *Get Quiz* use case, supports long-term writing improvement by reinforcing users' understanding of various writing elements and enhance their learning progress over time through interactive quizzes.

Requirement: This feature must generate relevant questions that cover the diverse range of writing elements. The system must also intelligently select appropriate questions from the question pool based on the user's learning needs, ensuring that each quiz remains engaging and educational. The quizzes should be user-friendly and short.

- **Review:** The review feature enabled through reflection, depicted by the *Get*

Review use case, is also part of the system's long-term writing improvement features. It allows users to review past work and receive feedback that highlights both strengths and areas for improvement. On the one hand, it includes the *Tips* feature for personalized suggestions, refining the user's skills and identifying recurring patterns in their writing. On the other hand, it includes the *User Improvements* inclusion for tracking progress over time offering insights into how their writing has evolved. The *Choose LLM* option is also available, allowing users to select the LLM for the generating the review.

Requirement: The review feature must provide useful feedback, offer personalized tips, track user progress accurately, and allow users to request reviews as needed.

Gamification

- **Gamification:** The gamification feature, depicted by the *Get Gamification Progress* use case, is designed to capture user engagement through elements such as XP, levels, badges, and achievements. XP and levels are automatically updated, while badges and achievements serve as optional extensions.

Requirement: To effectively support the idea of gamification, these elements must be thoughtfully designed and visually accessible. This includes using progress bars and clear labels placed consistently within the user interface (UI), such as on the user's profile page, ensuring that users can easily track their progress. As already mentioned previously, it is important to note that we only integrate gamification to track users activity.

3.1.2 Non-Functional Requirements

Non-functional requirements describe how a system should function, meaning how users perceive the final system e.g. in terms of performance and usability. We see the following non-functional requirements as essential for our system:

1. **Response Time:** The system should efficiently process requests, requiring optimization in handling large data volumes within a single request. To ensure quick response times and minimize latency for an enhanced user experience, it's essential to recognize that each component of the system, from data generation to processing, contributes to the overall response time. Therefore, it is crucial to select and optimize all system components, especially the LLM, to ensure the entire process remains responsive and fast.
2. **Response Quality:** The system must produce high-quality responses that are directly related to the functional requirements. These responses should effectively demonstrate the system's ability to understand the user's needs. They should be accurate and relevant to the user's academic writing progress and presented in a way that is easy to understand and can either be immediately applied by the user or contribute to their long-term improvement.

3 Concept

3. **Cost Efficiency:** The implementation of our system should not require large financial resources, also leading to the correct model selection.
4. **Model Selection:** Closely related to response time, quality, and cost efficiency is the selection of the model, which must be carefully chosen to achieve fast response times and maintain high output quality, directly impacting the system’s overall effectiveness. The model should be cost-efficient, delivering strong performance without exceeding budget limits. Additionally, it needs to be easy to integrate into our current system and adaptable for future updates.
5. **Design:** Another noteworthy non-functional requirement includes the design. For new users to understand and use the system without any difficulties, it is crucial that the design is simple and intuitive.
6. **Data Privacy:** Transparent data privacy regulations are essential to build trust with users and thus promotes a positive perception of the system. The system must clearly communicate how user data is collected and used, which is also attached to global data protection regulations.

3.2 System Design

Based on the requirement analysis, we developed WILLM which is particularly targeted at non-native English-speaking students who often struggle with academic writing. The tool is designed to work with input from a single paper, focusing on improving the writing quality within that context. Although our current work is not optimized for handling multiple papers, this could be explored in future work 6.2. WILLM is based on a client-server architecture and consists of multiple components, each playing its role for academic writing improvement based on LLMs. Figure 3.2 illustrates the architecture and its main components. These components are explained in the following sections, including their functions and purpose.

3.2.1 Components

Application Programming Interface

The application programming interface (API) acts as the central connection point between the system’s different parts. It manages the flow of information between the UI, the LLM, the user database, the embedding model, and the vector database. The API receives requests from the user, forwards them to the relevant components and returns the responses such as feedback, quizzes or reviews to the UI.

LLM

The LLM is the main component responsible for processing user inputs and generating content. It performs the following main tasks from the functional requirements:

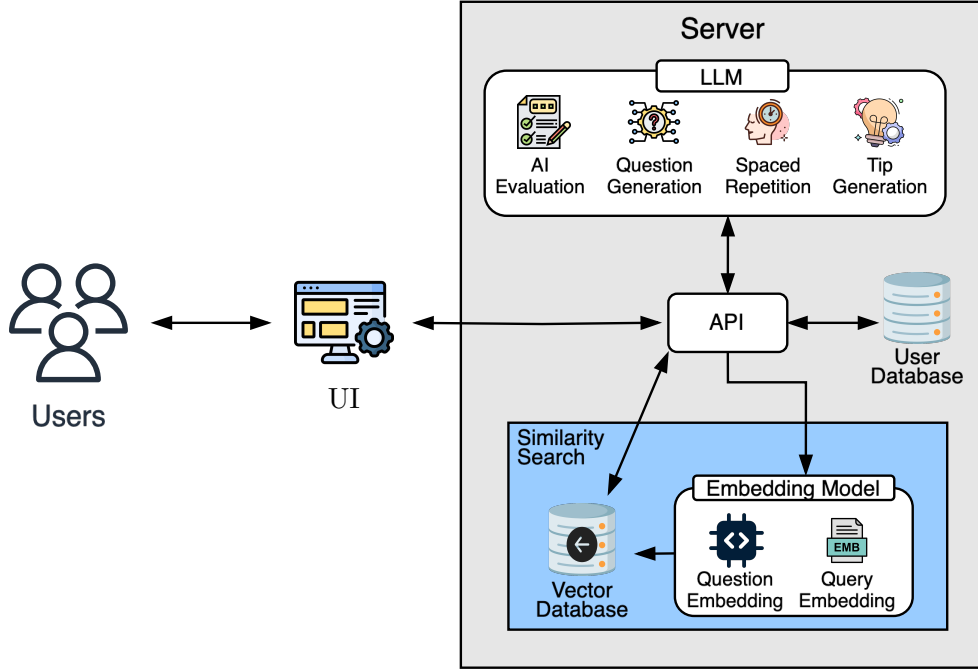


Figure 3.2: The proposed system overview of WILLM. Arrows denote an established communication channel between the components [Fla].

1. **AI Evaluation:** The LLM provides the corrections on user writing by analyzing aspects such as grammar, vocabulary, organization, coherence, and writing style and providing corresponding analytic scores. This helps users improve their writing in the short-term.
2. **Question Generation:** The LLM creates the quiz questions based on the user's submissions aimed at improving writing skills over time.
3. **Spaced Repetition:** The LLM uses spaced repetition to ensure long-term retention of writing skills. It reviews previous quiz results and selects which questions need to be included in future quizzes to reinforce learning by focusing on the topics that need the most improvement.
4. **Tip Generation:** The LLM generates personalized tips to help users improve specific areas of their writing, specifically on coherence and organization.

User Database

The user database stores all the information related to users, including their profiles, writing history, quiz scores, reviews, and other progress data. It also stores user preferences, such as selected language and model settings.

Similarity Search

The similarity search component in combination with the embedding model allows the system to find relevant matches for user inputs. The embedding model transforms both questions and user queries into vector representations, which are stored in the vector database. The system then uses them to compare current inputs with previously stored data, retrieving relevant results with similarity searches to provide appropriate quizzes.

User Interface

The UI is the main point where users interact with the backend. The UI displays the AI Evaluation, quizzes, reviews, and progress tracking in a simple and understandable way. In addition, it includes gamification elements, such as badges and progress indicators, that encourage users to stay motivated and engaged in improving their writing.

After having established the main components of WILLM, we now discuss the underlying processes of the immediate and long-term writing improvement features with sequence diagrams. To clarify the sequence diagrams and make them more relatable, we are going to explain them using a running user example. In this case, we follow Sarah, a user aiming to improve her academic writing skills using WILLM.

3.2.2 Immediate Writing Improvement

AI Evaluation

Sarah submits a section of her research paper draft to WILLM through the UI, looking for feedback on grammar and organization. After submitting her text, the following sequence of actions is initiated, as depicted in Figure 3.3.

1. **User Submission:** The system receives Sarah's draft via the UI, and the API immediately stores the submission in the user database.
2. **Generate Corrections:** The API sends a request to the LLM to evaluate Sarah's text. The LLM processes the draft, identifying grammar mistakes, unclear sentences, and providing explanations for each correction based on the language she selected. The API then performs post-processing to ensure the correction details are properly formatted and linked to Sarah's submission.
3. **Generate Scores:** The API requests the LLM to evaluate Sarah's submission against our custom writing rubric. The LLM generates analytic scores for each criterion and returns them to the API, where post-processing is performed to ensure the scores are consistent with the rubric.
4. **Display Feedback:** The API stores the complete evaluation including corrections and scores in the user database and returns the full AI Evaluation to Sarah via

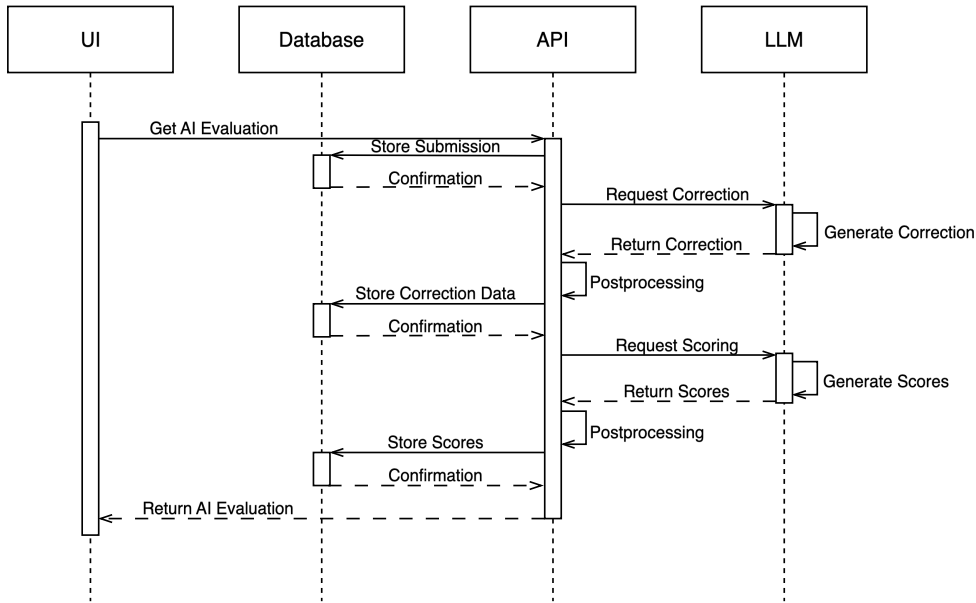


Figure 3.3: The sequence diagram depicting the AI evaluation of our system.

the UI. This evaluation includes detailed corrections, explanations, and scores across all relevant writing elements.

Thus, the AI evaluation process provides real-time feedback, allowing Sarah to quickly identify and address issues in her writing. This enables her to improve the quality of her draft immediately.

Rubric Design

One main feature of WILLM's writing evaluation process is the use of a custom-designed rubric for assessing users' writing submissions. Instead of relying solely on pre-existing rubrics such as TOEFL iBT [ETS], IELTS [IEL], or PTE Academic [PTE], we decided to design our own rubric that better fits the goals of our system. The reasoning behind this is that each of these rubrics emphasizes different writing elements, and not all of them align with the areas we prioritize for academic writing improvement. Therefore, we extracted and combined relevant criteria from each rubric, ensuring a comprehensive evaluation that addresses grammar, vocabulary, organization, coherence, and writing style. The full version of our rubric is provided in the Appendix A.

3.2.3 Long-Term Writing Improvement

After receiving immediate feedback on several drafts of her research paper, Sarah is now using WILLM's long-term improvement features. The system has tracked her progress over time and has begun generating questions that get included in quizzes to

3 Concept

help enhance her academic writing skills. Furthermore, Sarah can now request reviews that are generated based on the data WILLM collected through her submissions.

Question Generation

After every third submission Sarah makes through the UI, the system automatically generates a question to address recurring mistakes or areas that need attention. The sequence of actions, depicted in Figure 3.4, outlines the process for generating these questions:

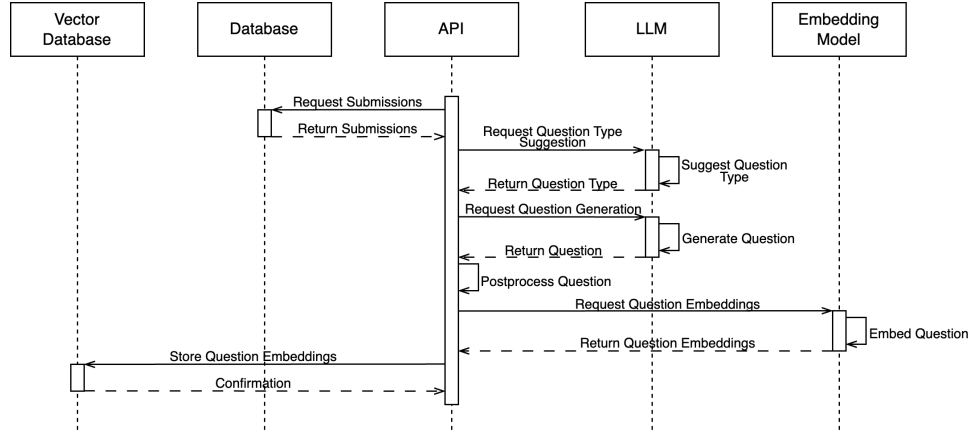


Figure 3.4: The sequence diagram depicting the question generation of our system.

1. **Request Submissions and Suggest Question Type:** The system collects Sarah's most recent submissions from the database and the LLM processes them to suggest a question type that targets the writing element needing the most improvement. For example, if Sarah consistently struggles with coherence, the system may suggest a question focused on improving coherence.
2. **Check Diversity:** To ensure variety, the system checks whether a particular question type has been overused. If necessary, the system selects a different question type.
3. **Question Generation:** The LLM eventually generates a specific question based on Sarah's mistakes, ensuring the content is relevant to her past submissions. This could involve asking Sarah to revise a grammatically incorrect sentence or identify a better synonym for a word she overuses.
4. **Post-Processing and Embedding:** The API post-processes the generated question to ensure it adheres to the predefined output structure, which includes the question, multiple-choice options, and/or the correct answer. This ensures the system can efficiently store the questions and handle a usable format for future quiz generation. The question is then embedded using the embedding model to ensure similarity searches can be performed during quiz generation.

5. **Storing in Question Pool:** The generated question is then saved in the question pool inside the vector database, where it can later be retrieved for quizzes.

Question Design

WILLM's question generation system is designed to cover the defined writing elements. After analyzing Sarah's submissions, the system generates a question from one of six question types. These are either textual-input (TI) or multiple-choice (MC) questions and are explained in the following:

1. **Revision (TI):** Sarah is given a sentence containing grammatical errors similar to those she has made in previous submissions. She is asked to revise the sentence and correct the mistake.
2. **Synonyms (MC):** A word from Sarah's submission is highlighted, and she must choose a more academic synonym from a set of five multiple-choice options. Only one option is the correct synonym.
3. **Organization (MC):** Sarah is asked to complete a sentence that ensures logical progression between two excerpts taken from different sections of her submission. This helps her understand the importance of structuring her writing cohesively.
4. **Coherence (MC):** A portion of Sarah's text is chosen, and she is asked to identify which of four multiple-choice sentences would best continue the passage in a coherent way.
5. **Academic Sentence (TI):** Sarah is asked to paraphrase a sentence from her submission to make it more formal and academically appropriate. This helps improve her academic writing style.
6. **Argument Strengthening (MC):** A weak argument in Sarah's submission is highlighted, and she must select the best option to strengthen it. This question type enhances her ability to construct well-supported arguments.

Quiz Generation

After generating a pool of questions based on Sarah's past submissions, the system is now ready to create quizzes. As Sarah continues to submit her drafts, WILLM uses her previous performance data to generate a personalized quiz. The sequence of actions for generating Sarah's quiz, depicted in Figure 3.5, is as follows:

1. **Query Generation:** The system collects Sarah's recent data, including mistakes and scores from her last three submissions, as well as her latest review. With this recent data, the LLM generates a query.
2. **Embedding and Similarity Search:** The generated query is embedded using the embedding model. The system performs a similarity search within the question

3 Concept

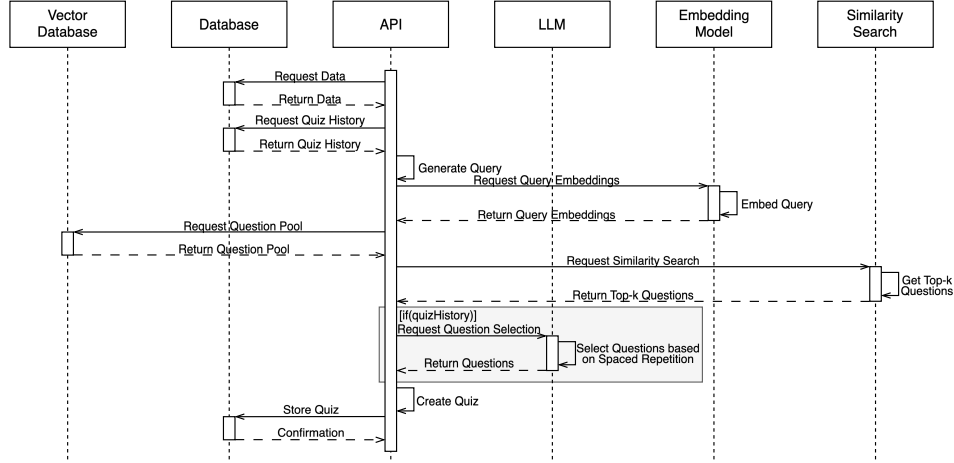


Figure 3.5: The sequence diagram depicting the quiz generation of our system.

pool to retrieve the top-k questions (where $k=10$ if quiz history exists, and $k=5$ without history) regarding the embedded query.

3. **Spaced Repetition:** If Sarah has taken previous quizzes, the system takes her quiz history and prompts the LLM to select the top five questions based on spaced repetition principles.
4. **Quiz Generation:** Once the questions are selected, the system compiles them into a quiz and stores it in the database. Sarah can then take this quiz to reinforce the skills she has been working on.

This quiz generation process ensures that Sarah receives questions that target her recurring mistakes and writing challenges, helping her improve her academic writing skills in the long-term.

Quiz Design

In each quiz, Sarah is presented with a question, and her answer leads to one of two possible outcomes:

- **Correct:** If Sarah answers correctly, the system marks the question as correct and Sarah can move on to the next question.
- **Incorrect:** If she provides an incorrect answer, the system will allow her to request an explanation of the correct answer. This triggers an additional request to the LLM to generate a detailed explanation. After viewing the explanation, Sarah can proceed to the next question.

This quiz design is based on research showing that feedback helps enhance the testing effect [BDKKM91] [KS89], regardless of whether the attempted answers are correct or

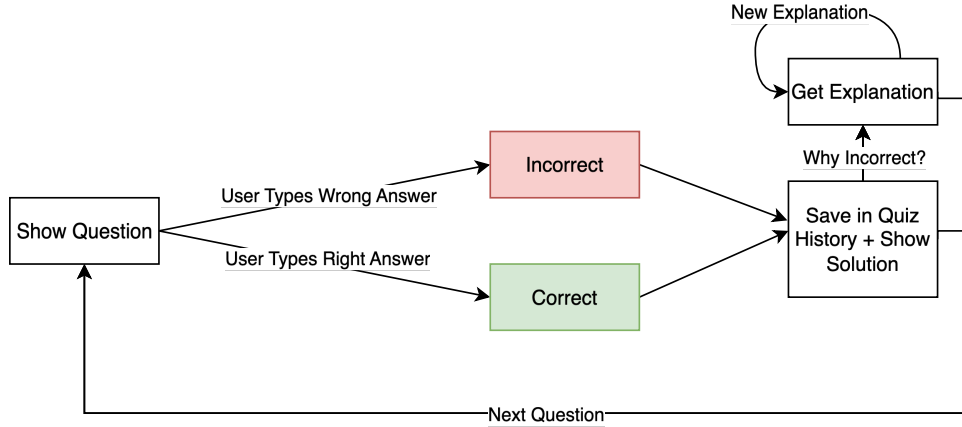


Figure 3.6: The diagram depicting the quiz design of our system.

not [BKRI08]. In any case, each answer is saved in the quiz history and the solution is displayed.

Quiz Schedules

Quizzes are introduced progressively over three weeks with intervals between quizzes gradually increasing to support spaced repetition principles. WILLM uses the following schedule:

- **First Quiz:** Presented within 24 hours of the first login.
- **Second Quiz:** Presented two days after the first quiz.
- **Third Quiz:** Presented four days after the second quiz.
- **Subsequent Quizzes:** Presented five days after the previous quiz.

If Sarah misses a quiz, the schedule reverts back to the previous interval. This ensures that Sarah stays on track without skipping important memory reinforcement steps. This schedule is designed to ensure optimal reinforcement of learning and is based on the work by Karpicke et al. [KB11] and Cepeda et al. [CVR⁺08] findings suggesting that testing intervals should be 20-25% of the total study period, with a progressive increase up to seven days.

Review Generation

In addition to quizzes, WILLM provides Sarah with long-term feedback through personalized reviews helping her assess her progress over time. These reviews consists of tips for improvement and highlights areas where Sarah has shown progress compared to her earlier submissions. The sequence of actions for generating Sarah's review, initiated by her request, is depicted in Figure 3.7 and involves the following steps:

3 Concept

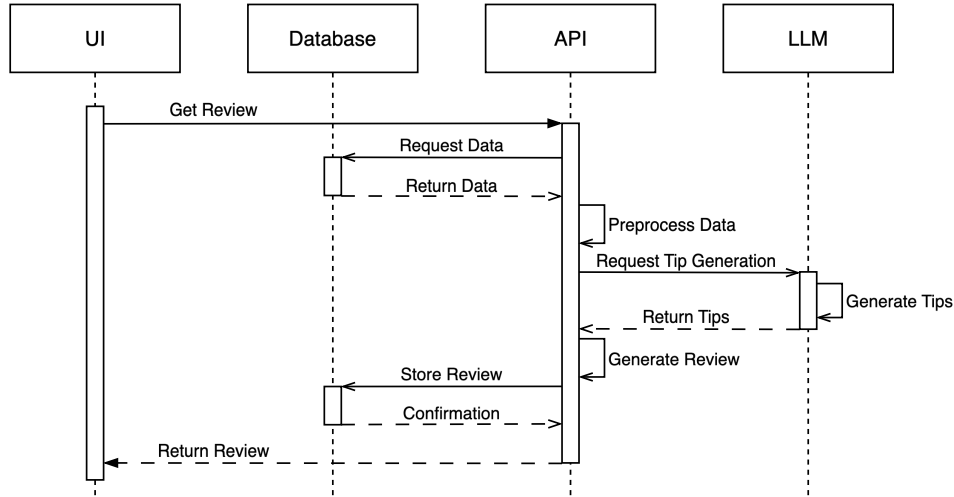


Figure 3.7: The sequence diagram depicting the review generation of our system.

1. **Review Request:** Sarah clicks the "Review" button in the UI, which prompts the system to begin the review generation process. This action is only enabled once she has submitted at least five pieces of writing.
2. **Tip Generation:** The API retrieves Sarah's last five submissions including the mistakes made and mistake types, also referred to as categories. This data is then pre-processed to calculate the frequency of mistakes in each category and targeted tips for these categories are created with the highest mistake frequencies, guiding the user toward the areas that require the most attention. Sarah can also receive additional tips for coherence and organization: If at least two sections for coherence and three sections for organization are identified in the user's submissions, a request is made to the API to analyze the relationships between the sections and create the tips. We chose coherence and organization to ensure users receive feedback not only within individual sections but also on how their writing flows between sections.
3. **Improvement Generation:** If at least ten submissions can be found, the system compares Sarah's most recent five submissions with her previous ten submissions. It identifies areas where her mistake frequency has decreased, marking these as improvements. This helps Sarah see where she has progressed over time.
4. **Review Generation:** The system compiles both the tips and the improvements. The review is stored in the database and presented to Sarah via the UI.

Score Comparison

As a smaller part of the review process, Sarah also receives a comparison of her recent scores with previous scores regarding the same sections. It allows her to see specific

areas where she has improved or needs further improvement.

3.2.4 Gamification

To track Sarah’s engagement with WILLM, the system’s gamification system rewards her with XP for daily logins, answering quiz questions correctly, and getting perfect scores in the quizzes. As Sarah accumulates XP, she progresses through levels, with each level requiring more XP than the last, up to a maximum level of 10. Sarah can also earn additional XP for maintaining a streak of daily logins or giving correct answers in quizzes. The build-in achievements further track her progress in areas like the number of quizzes completed and correct answers, and she can earn badges when meeting specific criteria. The detailed criteria for these achievements and badges, along with their corresponding rewards, are summarized in Appendix B, Table B.1.

3.3 Conceptual Limitations

After thoroughly tackling the requirement analysis and system design, we are going to discuss the known conceptual limitations of WILLM in this section. These limitations include issues related to the use of LLMs, the scope of the system for different user groups, and potential drawbacks in gamification elements.

3.3.1 Scope and Target Audience

Single Paper Restriction

WILLM’s testing period spans only three weeks, which limits the system’s ability to provide feedback on multiple papers. The short duration is tailored for users focused on improving a single academic paper, making it less suitable for users who are working on multiple projects.

Limitations for Advanced Users

The system is primarily designed to assist users with basic and intermediate academic writing skills. However, advanced users who are already strong in academic writing may find the system less helpful, as it is not optimized to provide complex writing critiques.

3.3.2 Limitations for LLMs

LLMs are a key component of WILLM, yet they come with inherent limitations that impact performance, scalability, and reliability. These limitations include computation time, resource costs, and hallucination.

Computation Time

Running larger LLMs requires significant computational power, leading to delays in generating responses. The complexity of tasks, such as evaluating long submissions or generating detailed feedback, often results in slower processing times, which can affect user experience, particularly for immediate writing improvement features.

Resource Costs

The costs associated with running larger LLMs at scale are high, especially when handling a large number of users simultaneously. Processing longer input texts or performing frequent evaluations places a heavy load on computational resources, which limits how the system can be realized.

Hallucination

One of the inherent risks of using LLMs is hallucination, where the model generates convincing but fabricated information. This can lead to situations where the LLM highlights mistakes that are not actually errors, especially in text that is already well-written in terms of academic standards. Additionally, the resulting explanations for these fabricated mistakes may also be inaccurate.

In conclusion, this chapter has outlined the conceptual framework and design principles behind WILLM. After discussing the functional and non-functional requirements that are necessary to build an efficient system, we provided a detailed explanation of the system's design including its architecture and main components. Finally, we addressed the conceptual limitations of WILLM. This basis sets up the next chapter, i.e. realization, in which we will examine the technical implementation of these concepts.

4 Realization

We now cover the exact implementation choices for WILLM by considering the design presented in Chapter 3.2. Then, in Section 4.1, we discuss the selection of appropriate tools and frameworks, explaining how the system components interact and outlining the main decisions and trade-offs made to optimize both functionality and user experience. Additionally, in Section 4.4, the limitations of the system are also addressed.

4.1 Architectural Overview

The system architecture is hosted on a server environment managed using Docker Compose, which organizes all necessary containers. The architecture is designed to provide an accessible platform for academic writing improvement based on LLMs. The following subsections detail the components involved and the tools chosen for each.

4.1.1 System Components

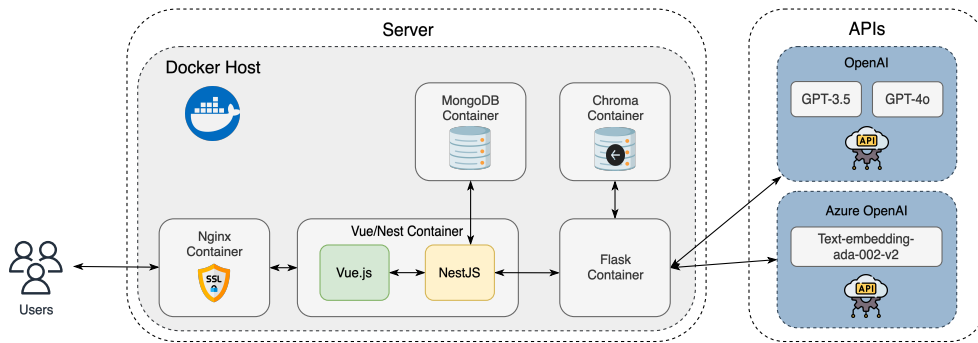


Figure 4.1: The architecture overview of WILLM [Fla].

The system's architecture comprises several components, each encapsulated within Docker containers. These components and the reasons behind their selection are as follows:

Ngix Container

Ngix is selected as the web server to handle incoming HTTP and HTTPS requests. It is chosen for its ability to manage SSL certificates effectively, ensuring secure com-

munication between users and the application. Nginx also redirects all HTTP requests to HTTPS and forwards these to the Vue/Nest container.

Vue/Nest Container

This container hosts both the frontend and backend of the application:

- **Frontend (Vue.js):** We choose Vue.js for the frontend development because of its efficient reactivity system, which allows for dynamic and responsive user interfaces. Vue.js is particularly well-suited for building single-page applications and its ease of use being an additional advantage.
- **Backend (NestJS):** NestJS on the other hand is selected for its robust, modular architecture that supports scalability and the microservice pattern, thus managing backend processes, including interactions with the MongoDB container and communication with the Flask API for tasks that require the LLMs or Chroma.

The combination of Vue.js and NestJS within a single container is a practical choice due to the NestJS build process, which incorporates a static build of the Vue.js frontend.

MongoDB Container

MongoDB is chosen for storing user-related data because of its NoSQL nature, which provides flexibility in handling dynamic and evolving data structures. This flexibility is particularly beneficial given the evolving nature of the thesis, as it allowed for easier restructuring of the database as requirements changed. In addition, MongoDB's document-based model is well-suited for managing unstructured or semi-structured data, which is often encountered in text processing applications. MongoDB stores several collections that are important for the operation of the application:

- **Issue:** Stores details about mistakes in user submissions, including the correction and the type of mistake.
- **Quiz:** Stores generated quizzes and user results.
- **Review:** Stores generated reviews based on user submissions.
- **Score:** Stores the scores assigned to user submissions.
- **Sessions:** Logs user sessions, with a new session generated each day after the user interacts with the system.
- **Text:** Stores user submissions.
- **User:** Contains user documents, including pre- and post-test submissions, gamification elements, and other related data.

Flask Container

The Flask API within the Flask container handles several essential tasks:

- **API Requests and Text Processing:** Flask processes API requests related to writing improvement, such as text correction, quiz and review generation. Its minimalistic design made it an ideal choice for rapid development, especially for the integration with Python-based tools like Langchain and ChromaDB.
- **Integration with External APIs:** Flask communicates with the Azure OpenAI API and OpenAI API to perform tasks like text embedding and language model processing. This integration is crucial for generating text embeddings, performing semantic analysis, and other AI-driven features.
- **Post-Processing and Vector Management:** After receiving data from external APIs, Flask performs necessary post-processing and manages vectorized data using Langchain, which powers features like similarity searches. While the functionality could have been developed without Langchain, its integration allows for future expansion, such as the potential incorporation of agents or assistants.

Flask is chosen for its simplicity and flexibility, particularly in handling API requests and interacting with Python-based libraries, making it a suitable choice for the backend tasks required by the system.

ChromaDB Container

ChromaDB stores and manages vectorized data, which is essential for tasks such as similarity searches. It is selected for its efficiency in handling large sets of vectorized data and its seamless integration with Langchain. This combination ensures that the system can perform complex vector similarity searches necessary for advanced writing improvement features.

External APIs

The system utilizes two external APIs:

- **OpenAI API:** Used for generating text embeddings, which are stored in ChromaDB. OpenAI is selected for their strong model performance in text embedding tasks.
- **Azure OpenAI API:** Provides access to language models for text generation and analysis. This API is mostly selected due to the ability to leverage the authentication provided by the chair of databases and information systems at RWTH University.

The architecture of this system is designed to provide an accessible platform. It leverages containerization through Docker Compose to support consistent and reliable deployment. The integration of the various components and external services is structured to create an effective tool for academic writing improvement using LLMs.

4.1.2 Model Selection

For WILLM, two primary language models were selected: GPT-3.5-turbo-1106 (subsequently referred to as GPT-3.5) and GPT-4o. These models were chosen for their performance, cost-effectiveness, and ease of integration with the system’s existing architecture. GPT-3.5 is chosen for tasks that require less complex language processing and reducing costs, while GPT-4o is selected for handling the opposite.

As embedding model, Text-embedding-ada-002-v2 is chosen. This choice was simple as the OpenAI API at that time didn’t offer any other for embedding models and we wanted to stay consistent regarding the API choice.

4.2 Prompt Realization

To meet the requirements discussed in Section 3.1, we focused on prompting techniques instead of fine-tuning models. As noted in related work 2.2, fine-tuning requires large datasets, meaning hundreds or thousands of examples [DCLT18], which are hard to obtain within the scope of this thesis. Recent research has also shifted towards prompt tuning [Kwo23], making it a more practical and cost-effective approach. We experimented with various prompting strategies to achieve high-quality responses while balancing performance and resource limitations.

Prompting Strategy

We decided to use zero-shot prompting throughout the system. This decision is taken primarily by considerations of token efficiency and cost optimization as explained in Section 4.4. With GPT-4o, the primary model used in the system, zero-shot prompting reliably generated structured and accurate outputs, thus minimizing the token usage and corresponding costs.

4.2.1 Immediate Writing Improvement Prompts

For immediate writing improvement, we devised prompts targeting specific areas such as grammar, vocabulary, organization, coherence, and writing style. The prompt for grammar and vocabulary correction is split into two versions. The first version, designed for GPT-4o, corrected grammar and vocabulary across the entire text, while the second version, used with GPT-3.5, worked at the sentence level. In the latter case, the text is divided into individual sentences, each processed separately for correction. This approach is necessitated by GPT-3.5’s performance limitations in handling full texts while maintaining the required output structure. The prompt designed for GPT-4o is depicted in Figure 4.2, and the sentence-level prompt for GPT-3.5 is shown in Figure 4.3. Note that we include the system prompt as well as the user prompt. Furthermore, the specific tokens in the text such as M, C, E, T, and X are used for processing the output from the LLM. After processing, the corrected version of the submission is passed through prompts assessing organization, coherence, and writing style. This ensures

that grammar and vocabulary errors were corrected before subsequent evaluations, preventing the model from overemphasizing them again.

To evaluate organization, coherence, and writing style, we employed a unified prompt that handled all three aspects simultaneously. Creating separate prompts for each element would have increased token usage and costs. Moreover, it risked generating redundant corrections, resulting in unnecessary complexity for users. Even handling these redundancies with extra requests would increase token consumption and costs. The system prompt and part 1 of the user prompt is shown in Figure 4.4 and part 2 of the user prompt is illustrated in Figure 4.5. This split was necessary due to the extensive length of the user prompt.

For scoring, we designed a single prompt that evaluated writing quality across the five writing elements. The structure of this prompt is illustrated in Figure 4.6. We removed the scoring rubric we developed due to its size; it can be found in the Appendix A in Table A.1.

4.2.2 Long-Term Writing Improvement Prompts

Turning to long-term writing improvement, we utilized different prompts for generating the questions for each type which are later included in the quiz. As explained in Section 3.2.3, there are six question types. The system prompt is shown in Figure 4.7. Revision is presented in Figure 4.8, synonyms in Figure 4.9 and organization in Figure 4.10. The prompt for the coherence question is illustrated in Figure 4.11 and for academic sentence in Figure 4.12. Lastly, argument strengthening is presented in Figure 4.13.

To generate quizzes, we needed the model to select questions based on the principles of spaced repetition. Instead of providing the full text of each question along with user responses, we simplified the process by only providing the question types and whether the user answered correctly or incorrectly. This decision minimized token usage and ensured the focus remained on improving areas where the user had shown weakness. The corresponding prompt can be found in Figure 4.14.

For reviews, which ensure that users receive feedback not only within individual sections but also on how their writing flows between sections, we needed to construct prompts for these scenarios. The system prompt is shown in Figure 4.15 and the user prompts are shown in Figure 4.16 and 4.17, respectively.

In summary, we employed a variety of prompt-based techniques to enhance both immediate and long-term writing improvements by focusing on zero-shot prompting and efficient token usage. The next section outlines the realization of the user interface.

SYSTEM PROMPT:

You are an assistant designed to help improve academic writing by providing detailed feedback on grammar and vocabulary. The user will submit a piece of writing, and your task is to identify and correct grammatical and vocabulary mistakes. Categorize each mistake into one of the specified categories. Provide the explanations in {language}.

USER PROMPT:

Regarding grammar and vocabulary, look for issues with misspelling, subject-verb agreement, tense consistency, pronoun agreement, incorrect use of articles, incorrect prepositions, inappropriate word choice, redundancy, and other grammatical errors. Provide the grammatical rule or reasoning when explaining the mistake. Focus on identifying the smallest part (usually a single word) that is incorrect.

Categorize each mistake into one of the following categories. If the mistake does not fit any of the categories, choose "Uncategorized":

- misspelling
- subject-verb agreement
- tense consistency
- pronoun agreement
- incorrect use of articles
- incorrect prepositions
- inappropriate word choice
- redundancy
- punctuation

For each mistake, you should provide the following:

1. The mistake: Highlight only the incorrect word or smallest possible segment that needs correction.
2. The correction: Provide only the corrected word or smallest possible segment.
3. The explanation: Explain why it is a mistake and provide the relevant rules or reasoning. Provide the explanations in {language}.
4. The category: Specify the category of the mistake.
5. The context: Provide exactly one word before and one word after the mistake for context. It must be exactly the mistake, not the correction.

Output the feedback in the following structure without wrapping it with quotation marks:

M: [Highlight only the incorrect word or smallest possible segment]

C: [Provide the corrected word or smallest possible segment]

E: [Explain why the grammar or vocabulary is problematic and how to improve it. Give the explanation in {language}]

T: [Category]

X: [Provide exactly one word before and one word after the mistake for context. It must be exactly the mistake, not the correction.]

Don't use bullet points or any other sort of list. Separate each set of mistakes/corrections/explanations/contexts with a blank line. If the submitted writing is good as it is, simply state: "The submitted writing is fine." Provide the corrected version of the text with the necessary changes after "Correction:".

Don't add anything else to the output. Now, correct the following submitted writing: {text}.

Figure 4.2: The system and user prompt for the correction of grammar and vocabulary issues with GPT-4o.

SYSTEM PROMPT:

You are an assistant designed to help improve academic writing by providing detailed feedback on grammar and vocabulary. The user will submit a sentence, and your task is to identify and correct grammatical and vocabulary mistakes. Categorize each mistake into one of the specified categories. Provide the explanations in {language}.

USER PROMPT:

Regarding grammar and vocabulary, look for issues with misspelling, subject-verb agreement, tense consistency, pronoun agreement, incorrect use of articles, incorrect prepositions, inappropriate word choice, redundancy, and other grammatical errors. Provide the grammatical rule or reasoning when explaining the mistake. Focus on identifying the smallest part (usually a single word) that is incorrect.

Categorize each mistake into one of the following categories. If the mistake does not fit any of the categories, choose "Uncategorized":

- misspelling
- subject-verb agreement
- tense consistency
- pronoun agreement
- incorrect use of articles
- incorrect prepositions
- inappropriate word choice
- redundancy
- punctuation

For each mistake, you should provide the following:

1. The mistake
2. The correction
3. The explanation in {language}
4. The category
5. The context

Output the feedback in the following structure without wrapping it with quotation marks nor brackets:

M: [Highlight only the incorrect word or smallest possible segment]

C: [Provide the corrected word or smallest possible segment]

E: [Explain why it is a mistake and provide the relevant rules or reasoning and how to improve it. Give the explanation in {language}]

T: [Specify the category of the mistake]

X: [Provide exactly one word before and one word after the mistake for context. It must be exactly the mistake, not the correction or a synonym.]

Don't use bullet points or any other sort of list. Separate each set of mistakes/corrections/explanations/categories/contexts with a blank line.

If the submitted writing is good as it is, simply state: "The submitted writing is fine."

Provide the corrected version of the sentence with the necessary changes after "Correction:". Don't add anything else to the output.

Now, correct the following submitted writing: {text}.

Figure 4.3: The system and user prompt for correcting grammar and vocabulary issues with GPT-3.5 on sentence level.

SYSTEM PROMPT:

You are an assistant designed to help improve academic writing by providing detailed feedback on organization, coherence, and writing style. The user will submit a piece of writing, and your task is to identify weaknesses and provide improvements. Categorize each mistake into one of the specified categories. Provide the explanations in {language}.

USER PROMPT:

Regarding organization: Identify problems with the logical flow of ideas, such as sudden shifts in topic, redundancy, or deviations from the main topic. Categorize each mistake into one of the following categories:

- disorganized ideas
- poor paragraph structure

Regarding coherence: Point out issues that affect the overall coherence of the writing, such as unclear references or lack of logical connections between sentences or paragraphs. Categorize each mistake into one of the following categories:

- irrelevant content
- poor logical flow
- poor transitions
- repetitive information

Regarding writing style: Suggest improvements related to writing style, including the use of active/passive voice, formal tone, clarity, and conciseness. Highlight only the words or phrases that need stylistic improvement and explain why the suggested style is preferable. Categorize each mistake into one of the following categories:

- formal tone missing
- missing precision and clarity
- passive voice overuse

Figure 4.4: The system prompt and part 1 of the user prompt for the correction of organization, coherence, and writing style issues.

Steps:

- Use formal English only.
- Tend to use common and easy-to-understand words or phrases.
- Avoid wordy sentences.
- Avoid using the same words or phrases repeatedly.
- Ignore grammar or vocabulary mistakes.
- Specifically focus on the section section.
- For each mistake, you should provide the following:
 1. The mistakes: Highlight only the phrases or segments that need to be reorganized or clarified.
 2. The corrections: Provide the corrected organization.
 3. The explanations: In less than 50 words, explain why the organization is problematic and how to improve it. Provide the explanations in {language}.
 4. The category: Specify the category of the mistake.

Output the feedback in the following structure:

Organization:

M: [Highlight only the incorrect phrase/segment in English]

C: [Provide the corrected phrase/segment in English]

E: [Explain why the organization is problematic and how to improve it. Give the explanation in {language}]

T: [Category in English]

Coherence:

M: [Highlight only the incorrect phrase/segment in English]

C: [Provide the corrected phrase/segment in English]

E: [Explain why the organization is problematic and how to improve it. Give the explanation in {language}]

T: [Category in English]

WritingStyle:

M: [Highlight only the incorrect phrase/segment in English]

C: [Provide the corrected phrase/segment in English]

E: [Explain why the organization is problematic and how to improve it. Give the explanation in {language}]

T: [Category in English]

If the submitted writing is good as it is for either organization, coherence or writing style, then the output should look like this: [Organization, Coherence, or WritingStyle]: The submitted writing is fine. Don't use bullet points or any other sort of list. Separate each set of mistakes/corrections/explanations/category with a blank line. Don't add anything else to the output.

Now, correct the following submitted writing: {text}

Figure 4.5: Part 2 of the user prompt for the correction of organization, coherence, and writing style issues.

SYSTEM PROMPT:

You are an assistant designed to evaluate academic writing. Your task is to assess the provided text and assign scores in five categories: grammar, vocabulary, organization, coherence, and writing style. Use the criteria based on IELTS, TOEFL, and PTE scoring rubrics. The scores should range from 1 to 9, with 9 being the highest proficiency level. Return the scores in the following format and no other:

Grammar: [Provide the score]

Vocabulary: [Provide the score]

Organization: [Provide the score]

Coherence: [Provide the score]

Writing Style: [Provide the score]

Ensure the output is properly formatted and includes all necessary keys.

USER PROMPT:

Please evaluate the following academic text and provide a score for each of the following categories: grammar, vocabulary, organization, coherence, and writing style. Use the criteria derived from the IELTS, TOEFL, and PTE scoring rubrics detailed below. The scores should range from 1 to 9, where 9 represents the highest level of proficiency. Return the scores in the following format and no other:

Grammar: [Provide the score]

Vocabulary: [Provide the score]

Organization: [Provide the score]

Coherence: [Provide the score]

Writing Style: [Provide the score]

Text:

{text}

Scoring Criteria:

... (Writing Rubric, see Appendix A)

Figure 4.6: The system and user prompt for the assessment of the five writing elements through scoring.

SYSTEM PROMPT:

You are an expert in academic writing. Your task is to generate various types of questions for academic writing improvement. Each question type has a specific format and requirement. Please follow the guidelines below to create each question type.

Figure 4.7: The system prompt for the generation of the question types.

USER PROMPT:

I will provide you with a text that includes a submission from a user. Identify the grammatical errors in the submission and generate a revision type question based on these errors. If there are none, simply generate a text with grammar mistakes. The text should have grammatical errors with no two possible answers to correct them. The text should be max 25 words. Provide the question and the corresponding correct text.

Text:

{lastSubmission}

Output format:

Type: revision

Question: Revise the text to correct the grammatical errors.

Text: [The text with errors]

Answer: [The correct text]

Figure 4.8: The user prompt for the generation of the revision question.

USER PROMPT:

I will provide you with a text that includes a submission from a user. Identify a word in the submission that could be replaced with an academic synonym and generate a multiple choice question for identifying academic synonyms. Provide the word and five options, with only one correct synonym. The difficulty should be medium.

Text:

{lastSubmission}

Output format:

Type: synonyms

Question: Identify the academic synonym for the word.

Word: [The word]

Options: [The options enumerated with A, B, C, D, E]

Answer: [The correct synonym]

Figure 4.9: The user prompt for the generation of the synonyms question.

USER PROMPT:

Given the provided text which includes two excerpts from sections (Introduction, Literature Review etc.) of a user his academic paper, create one multiple-choice sentence completion question to test the user's understanding of logical progression in academic writing.

Text:

{text}

Scenario: Review the first excerpt and extract a key point or section transition from the first excerpt. The scenario should reflect a common theme or recurring topic from these submissions.

Question: Complete the sentence to ensure a logical progression.

Options: Provide four options based on the second excerpt for how the next section might logically continue the discussion based on the first excerpt. The options should be generated so the correct answer is not too hard to find.

Answer: Indicate the correct option that best follows the scenario in a logically coherent manner.

Output format:

Type: organization

Question: Complete the sentence to ensure a logical progression.

Scenario: [Generated scenario description based on one of the last three user submissions]

Options: [Four options enumerated with A, B, C, D]

Answer: [The correct option]

Figure 4.10: The user prompt for the generation of the organization question.

USER PROMPT:

Given the provided text which includes excerpts from sections (Introduction, Literature Review etc.) of a user his academic paper, create one multiple-choice sentence completion question to test the user's understanding of coherence and logical flow in academic writing.

Text:

{text}

Scenario: Review the excerpts and extract a finding or statement from one of the texts that requires a follow-up sentence for improved coherence. The scenario should reflect a common theme or recurring topic from these submissions.

Question: Which sentence best follows for coherence?

Options: Provide four options for sentences that could logically follow the given scenario based on the context of the user's submissions. The options should be generated so the correct answer is not too hard to find.

Answer: Indicate the correct option that best maintains coherence and logical flow in the context of the user's submissions.

Output format:

Type: coherence

Question: Which sentence best follows for coherence?

Scenario: [Generated scenario description based on one of the last three user submissions]

Options: [Four options enumerated with A, B, C, D]

Answer: [The correct option]

Figure 4.11: The user prompt for the generation of the coherence question.

USER PROMPT:

I will provide you with a text that includes a submission from a user. Identify a sentence in the submission that needs to be paraphrased into academic style and generate a question based on this sentence. Provide only the original sentence of it. If the original sentence is longer than 30 words, please cut it down to 30 words by keeping the meaning.

Text:

{lastSubmission}

Output format:

Type: academic_sentence

Question: Paraphrase the following sentence into academic style

Sentence: [The original sentence]

Figure 4.12: The user prompt for the generation of the academic sentence question.

USER PROMPT:

I will provide you with a text that includes a submission from a user. Identify an argument in the submission that could be strengthened and generate a multiple choice question to strengthen the academic argument. Provide the argument and five options, with only one correct option to strengthen the argument. The argument and options should be maximum 25 words each.

Text:

{lastSubmission}

Output format:

Type: argument_strengthening

Question: Strengthen the argument by selecting the most appropriate option.

Argument: [The argument that needs to be strengthened]

Options: [The options enumerated with A, B, C, D, E]

Answer: [The correct option]

Figure 4.13: The user prompt for the generation of the argument strengthening question.

SYSTEM PROMPT:

You are an expert in spaced repetition and educational testing.

USER PROMPT:

You will be provided with a set of new questions and the history of the last two quizzes taken by a user. Your task is to return five question ids based on spaced repetition principles and add them to an array. Each new question only contains the question id and the question type. Each quiz history includes the question id, question type and whether the user answered correctly (result: true) or incorrectly (result: false).

Your task is to prioritize questions that the user got wrong previously when deciding which questions to include in the next quiz. Additionally, ensure there is a balance across different question types and sources (the quiz history and new questions). For example, if the user got a question type wrong more than once, that question type should have the highest priority.

Here are the new questions:

{new_questions}

Here is the quiz history:

{quiz_history}

Based on spaced repetition principles, return the five best question ids from the new questions for the next quiz in the following array output format:

Question IDs: [The five question ids separated by commas]

Don't add anything else, such as explanations or additional comments.

Figure 4.14: The system and user prompt for the question selection based on spaced repetition principles.

SYSTEM PROMPT:

You are an expert in academic writing, coherence and organization. I will provide you with text that includes excerpts from sections (e.g., Introduction or Methodology) of a user's last submissions. Your task is to generate a tip of maximum two sentences that helps the user improve either coherence or organization between these sections.

Figure 4.15: The system prompt for the generation of organization or coherence tips.

USER PROMPT:

Generate a tip of maximum two sentences that helps the user improve the coherence.

Text:

{text}

Output format:

Tip: [The tip to improve coherence]

Figure 4.16: The user prompt for the generation of organization tips.

USER PROMPT:

Generate a tip of maximum two sentences that helps the user improve the organization.

Text:

{text}

Output format:

Tip: [The tip to improve organization]

Figure 4.17: The user prompt for the generation of coherence tips.

4.3 User Interface

The user interface of WILLM is designed to give an intuitive and user-friendly experience for users to improve their academic writing skills. After registering and logging in, users can access all the features described in Section 3.1.1, functional requirements.

4.3.1 Immediate Writing Improvement Features

Upon logging in, users are presented with a dashboard as shown in Figure 4.18 where they can submit their texts for evaluation (1) along with the corresponding section (2). Once a text is submitted, users can choose between different learning modes (3): In productive mode, users can click on highlighted mistakes, such as grammar and vocabulary errors, to automatically apply the correction. In learning mode, users are prompted through a modal to manually correct the mistakes before applying the correction. Mistakes (4) are highlighted in red, and hovering over them reveals the mistake type, correction, and an explanation in the user's chosen language. Furthermore, scores from our scoring system (5) along with feedback on organization, coherence, and writing style (6) appear on the right side of the dashboard. For additional help, information icons (7) are integrated throughout the interface. Users can hover over these icons to get information on using WILLM.

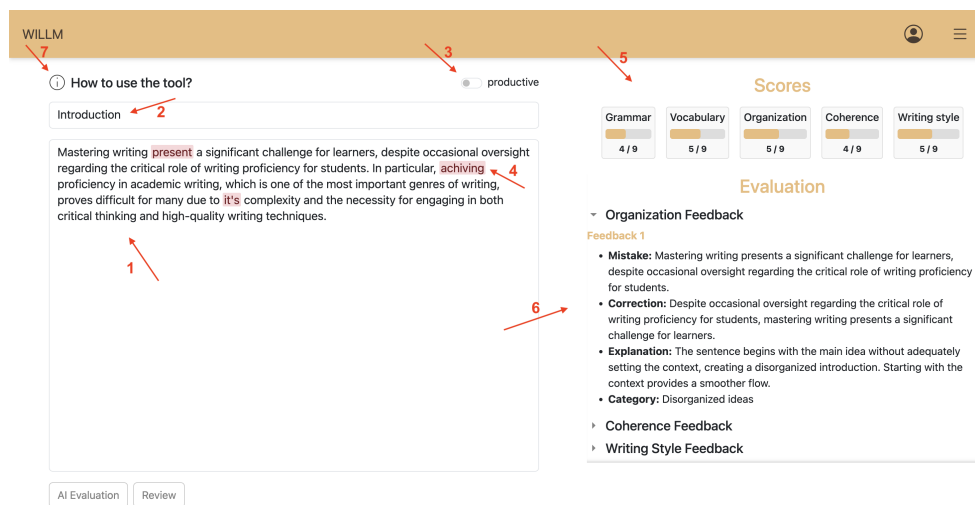


Figure 4.18: An example of how the text evaluation could look like after clicking on the button "AI Evaluation".

4.3.2 Long-Term Writing Improvement Features

Review

By clicking the Review button (8) depicted in Figure 4.19, users can generate a review based on their recent submissions, where they are provided with personalized feedback. This review includes personalized tips (9) generated by the LLM to target areas of improvement along with improvements (10) that highlight where the user has made progress.

After navigating to their profile page through the hamburger menu (11), users can view their scoring comparison (12). This feature shows the median of all previous scores for a selected section and compares it against the most recent score. Improvements are marked in green while declines are shown in red, providing a clear visual review of the user's performance over time.

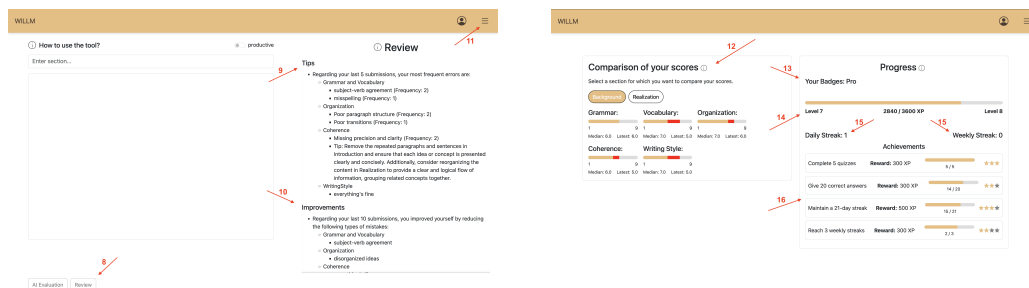


Figure 4.19: (left) An example of how the review could look like after clicking on the button "Review". (right) An example of a user profile including the score comparison on the left, and the gamification progress on the right.

Quiz

At certain intervals, as explained in Section 3.2.3, users are presented with a quiz. Quizzes can be accessed by navigating to the corresponding page through the hamburger menu. An example question is presented in Figure 4.20, where the question (17) appears at the top, followed by additional instructions (18). For textual-input questions, a text-area is presented and for multiple-choice questions, the answer options are interactively presented at the bottom (19).

4.3.3 Gamification

Additionally, Figure 4.19 also illustrates how users can see their detailed gamification progress. The gamification progress shows their current badge (13), level (14), and the status of their daily or weekly streak (15). The interface also displays the user's achievements (16) along with the corresponding rewards at each stage.

Finally, users can access a summary of their overall progress (20) and some settings by clicking on their profile icon in Figure 4.19, which brings up Figure 4.21. Here, they

WILLM Quiz

Question: Which sentence best follows for coherence?

Scenario: Tools like Grammarly, ProWritingAid, and Quillbot exemplify this by analyzing the tone and style of writing and offering nuanced corrections and suggestions tailored to the context, such as academic writing. Since NLP techniques are becoming more popular these days, research also focuses on leveraging LLMs - a subset of NLP - for this task. However, despite the potential of LLMs, there are only a few writing improvement tools using them as discussed in Section 3, related work. LLMs are advanced artificial intelligence systems designed to understand, generate, and manipulate human language. These models are trained on vast amounts of text data, allowing them to grasp the context and complexities of language.

Options:

- A. As a result, these writing tools offer more personalized suggestions.
- B. Therefore, it is important to continue this line of research to better understand LLMs.
- C. Many students have found these tools useful for improving their essays.
- D. Despite their advanced capabilities, the integration of LLMs in writing tools remains limited.**

Figure 4.20: An example of how a question could look like inside the quiz.

can set preferences such as the explanation language (21), the model they want to use for certain tasks (22) and check how many GPT-4o requests are remaining for the day (23).

User: benthillen

Your Progress

Lvl 6 2450 XP Lvl 7

Daily Streak: 1 Weekly Streak: 0

Language Settings

Explanations in: English

Model Settings

Grammar/Vocabulary: gpt-4o

Organization/Coherence/Writing Style: gpt-3.5-turbo

Score: gpt-3.5-turbo

Review: gpt-3.5-turbo

Requests left for gpt-4o: 12

Figure 4.21: An example of how the user profile looks like.

4.4 Decisions and Limitations

In this section, we discuss further decisions made during the realization of WILLM, the trade-offs involved, and the limitations faced in implementation. These decisions were influenced by the need for balancing system performance, cost-efficiency, and the quality of the responses generated by the system.

4.4.1 Model Choice for Users

To balance cost and functionality, users can adjust models for some features while others are fixed to ensure quality.

- **User-Controlled Features:** For tasks like grammar correction and scoring, users can choose between GPT-3.5 and GPT-4o. By giving users control over these features, they can manage costs while adjusting the model as needed. Default models are set primarily to GPT-3.5 to optimize expenses.
- **Fixed Features:** For critical tasks like question generation and spaced repetition-based question selection, GPT-4o is fixed. These tasks are complex and essential for long-term improvement, and the higher-quality output from GPT-4o ensures system effectiveness. Users cannot change models for these tasks.

4.4.2 Cost Optimization

To manage the high costs associated with running large models like GPT-4o, we implemented several strategies:

- **Input Length Restriction:** Users are restricted in how much text they can submit at once (500 words). This helps control token usage, especially when processing requests with GPT-4o, which is more expensive per token.
- **GPT-4o Request Limit:** Users are allowed up to 15 requests per day with GPT-4o, and the UI provides clear feedback on how many GPT-4o requests remain. This helps users plan their usage of the higher-quality model without exceeding daily limits.
- **Unlimited GPT-3.5 Requests:** To compensate for the limited access to GPT-4o, users can make unlimited requests using GPT-3.5. However, due to GPT-3.5's weaker performance in complex tasks, this occasionally led to incomplete or incorrect output structures, causing issues in post-processing on the backend. This could have been handled by a stronger model, GPT-4o mini, which was only released during the testing period. This model would likely have provided more consistent results, without increasing the overall cost of running WILLM.

By implementing these trade-offs and decisions, we were able to ensure that users can continue using the system without incurring high costs, but at the risk of receiving lower-quality feedback.

4.4.3 Testing Constraints

Due to time constraints, we were unable to do extensive testing of quiz generation and question design. This limited testing could affect the results of the system's long-term feedback features.

4 Realization

This chapter discussed the realization and implementation of WILLM, providing an overview of the system architecture and the tools and technologies used. We discussed how the various components interact within our environment and addressed the decisions and trade-offs made to especially manage the costs, including the restriction of GPT-4o usage and reliance on GPT-3.5 for less complex tasks. Limitations were also presented. Moving forward, the next chapter will cover the evaluation of WILLM and its effectiveness in improving academic writing skills.

5 Evaluation

In this chapter, we present a comprehensive evaluation of WILLM, focusing on the effectiveness of its core features in enhancing users' academic writing skills. Section 5.1 describes the methodology of the user study, including the pre- and post-test design, the evaluation of the main writing elements, and the tools used to gather data on user performance and engagement. Section 5.2 presents the results of the pre- and post-test evaluations, including statistical analysis of writing improvements, the effectiveness of quizzes, reviews, and feedback, and the usability of the system. The limitations of the evaluation are outlined in Section 5.4. Finally, Section 5.3 discusses our findings and the impact of WILLM's features on academic writing.

5.1 Methodology

This section covers the methodology by using a user study with the goal of evaluating the effectiveness of various features of WILLM. As already explained, the tool is particularly targeted at students who are in the process of developing their academic writing skills, rather than those who are already advanced in this area. The study aims to measure the impact of these features on the participants' writing abilities over time, assessing both immediate improvements and long-term retention.

5.1.1 Study Design

The study follows a within-subjects design, where all participants are exposed to the same intervention and their performance is compared before and after using the tool. It is an observational study meaning that participants were not randomly assigned to different conditions or interventions but rather all were exposed to the same tool. The primary focus is on assessing changes in participants' academic writing abilities and their perceptions of the tool's usability over time. Therefore, the study employs a pre- and post-test design over a three-week period to assess the effectiveness of the academic writing improvement tool. Additionally, a system usability scale (SUS) will be administered to gather participants' feedback on the usability of the tool, hoping that the combination of those two evaluation methods will provide a comprehensive evaluation of the tool's effectiveness and user experience.

Pre- and Post-Test

Participants begin the study by completing a pre-test to establish a baseline measure of their academic writing skills and conclude with a post-test, which mirrors the con-

ditions of the pre-test for comparability. Both tests require participants to submit a text of 300 words $\pm 10\%$, following specific guidelines to ensure consistency. After the three-week intervention period, where they accessed and interacted with the various writing features of WILLM, participants will submit a post-test text on the same topic or section as in the pre-test. They may submit up to three different writing samples, which can include sections from a research paper (e.g. introduction or methodology), academic essays, or reports. To ensure that the evaluation accurately reflects the participants' writing abilities, they must produce these samples independently, without assistance from any writing improvement tools.

System Usability Scale

After the three-week intervention period, participants will also complete a SUS to provide feedback on the tool's usability. SUS is a widely used tool in user experience research to assess the overall usability of a system. Since its first development by John Brooke in 1986 [Bro96], SUS has been the industry standard for assessing a broad range of products. The scale consists of ten questions, with responses recorded on a five-point Likert scale ranging from "Strongly Disagree" to "Strongly Agree." The switching positive and negative phrasing of the questions helps reduce the response bias and provides a more balanced view of the system's usability [Lew18]. To calculate the SUS score, each item's raw score is adjusted. For positively worded items, the score is reduced by 1. For negatively worded items, the score is subtracted from 5. These adjusted scores are then summed to get the total score, which is multiplied by 2.5. This process converts the score to a range of 0 to 100, where higher scores indicate better usability [Lew18].

5.1.2 Participants

We recruited 19 participants (5 females, 14 males), comprising 5 undergraduates, 9 Bachelor's degree and 5 Master's degree students. Selection was based on the following criteria:

1. **Education Level:** Participants included undergraduates (excluding first-year students) and graduate students with Bachelor's or Master's degrees, ensuring academic proficiency relevant to the study.
2. **Language Proficiency:** Non-native English speakers were chosen to assess the tool's effectiveness for users who may face additional challenges in academic writing due to language barriers.
3. **Prior Experience:** The participants need to have prior experience with writing tools and LLMs to ensure they could effectively engage with and evaluate the tool's features.

Ethical Guidelines

All participants were informed about the data privacy protection, access to their personal data and their rights. This information was provided through a consent form which participants were required to fill out and sign before participating in the study.

5.2 Results

In the following sections, we present the results of the pre- and post-test evaluations, along with the analysis of our features for enhancing users' academic writing skills. This includes the immediate and long-term writing improvement features, the impact of the integrated gamification, and WILLMs' usability.

5.2.1 Immediate Writing Improvement

One of the primary goals of this thesis is to assess the immediate impact of WILLMs' feedback system on users' writing performance. In this section, we focus on the immediate improvements observed after participants received feedback, and how this feedback influenced their performance in the different writing elements.

Submission Rates, Feedback Frequency, and Score Distribution

To begin the analysis of immediate improvements, we begin by analyzing the submission rates. This helps contextualizing the upcoming analyses in subsequent sections. The submission statistics over our three-week testing period are summarized in Table 5.1 below, where all submissions were corrected with GPT-4o and scored with GPT-3.5, as users didn't change the default settings.

The submission data shows a wide range of engagement levels, with a mean submission rate of 14.69 per week and an overall average of 23.32 submissions. While some users submitted as few as 7 pieces over the study period, others submitted up to 95. This provides context for the following feedback analysis.

Metric	Weekly	Overall
Mean	14.69	23.32
Median	15.75	16.0
Minimum	3.50	7.0
Maximum	39.12	95.0
Standard deviation	7.89	20.26

Table 5.1: Summary of submissions.

We continue by examining how frequently feedback was provided, measured by the average number of identified issues across different writing scores for each element, as shown in Figure 5.1. This analysis helps to determine whether users with lower scores

5 Evaluation

received more detailed feedback and if those interventions were associated with greater improvements in their writing performance, as examined in a subsequent part. The number of submissions is also provided to give context to the feedback data, giving a better understanding of how many submissions contributed to the average number of issues identified. Score 1 was excluded from this analysis due to no available data.

The results show that for grammar, as scores increase, the average number of identified issues decreases with the exception of score 3. For vocabulary, the feedback generally decreases from a score of 4 to a score of 9, except for score 2 and 3 and an outlier at score 6. However, this pattern does not hold for all writing elements. In writing style, feedback generally decreases as scores move from medium to higher but the range stays between 0 and 1. In organization and coherence, feedback for lower scores is sometimes less frequent than for medium scores.

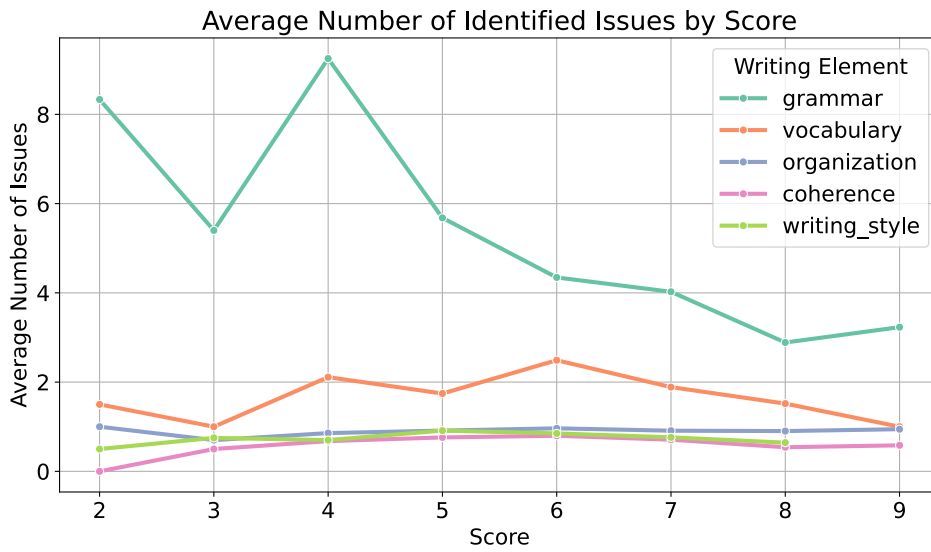


Figure 5.1: Average number of identified issues per writing element by score.

Next, we explore the score distribution for each writing element. Figure 5.2 visualizes the distribution of scores for each writing element by using a violin plot [Eta24]. This plot highlights the density and spread of scores across all submissions, showing how the scores cluster around certain values for each writing element.

By analyzing the plot, we notice that users tend to perform better for grammar and organization as indicated by the thicker areas at higher score ranges. The more uniform spread of scores for writing style and the lower upper quartile of 7 indicate that users had more varied performance in this element. Scores for vocabulary and coherence are more tightly clustered around their median values. Additionally, the median score for most elements is 7 whereas the lower quartile across all elements is 6.

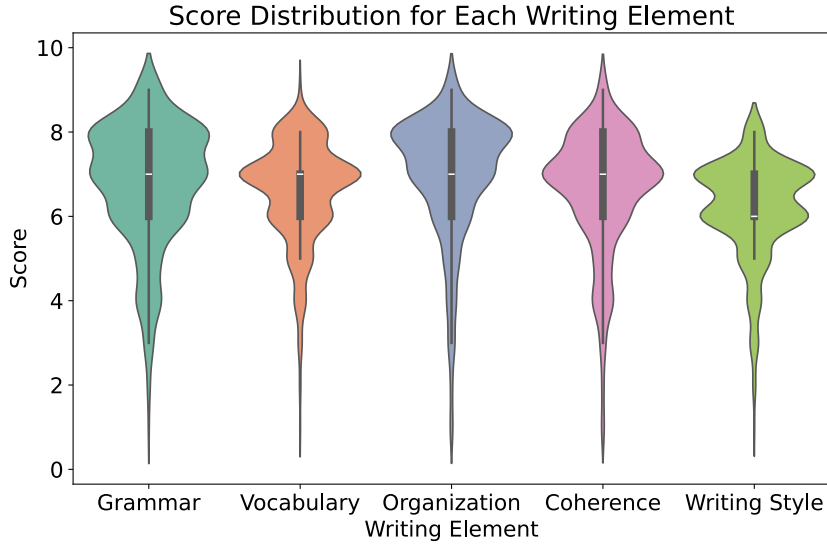


Figure 5.2: Score distribution for each writing element. The bar in the center of each violin represents the interquartile range, which includes 50% of the scores and the small line shows the median score.

Pre- and Post-Test Data

To analyze the improvement of the user from pre- to post-test submissions, each user’s writing was evaluated across our five writing elements. As demonstrated by Mizumoto et al. [ME23], LLMs can effectively function as AES tools. To enhance consistency in the evaluation process, we scored each pre- and post-test five times using GPT-3.5 using our writing rubric and calculated the mean to provide a final score for each element [SKC⁺22] [TSB⁺24].

Descriptive statistics for each writing element in both the pre- and post-tests are presented in Table 5.2. These values represent the overall performance trends across the participants.

As observed in Table 5.2, there is a general increase in the mean scores from the pre- to post-test across all writing elements. For example, the mean score for grammar increased from 6.22 to 6.76, and coherence improved from 6.68 to 7.10. The standard deviation in these metrics indicates how much variation or dispersion exists from the average. The reduction in standard deviation for most elements, such as grammar from 0.94 to 0.65 and coherence from 0.84 to 0.70, indicates a tighter clustering of scores around the mean in the post-test, suggesting greater consistency in performance across participants.

In addition to the overall scores, the types of mistakes encountered during the pre- and post-test phases provide more specific insights into how writing improved. Only

Writing Element	Test	Mean	Median	Std.
Grammar	Pre-Test	6.22	6.40	0.94
	Post-Test	6.76	6.90	0.65
Vocabulary	Pre-Test	6.35	6.40	0.71
	Post-Test	6.72	6.60	0.70
Organization	Pre-Test	7.11	7.20	0.73
	Post-Test	7.47	7.60	0.66
Coherence	Pre-Test	6.68	6.80	0.84
	Post-Test	7.10	7.20	0.70
Writing Style	Pre-Test	6.07	6.13	0.76
	Post-Test	6.45	6.50	0.60

Table 5.2: Descriptive statistics for pre- and post-test scores.

grammar and vocabulary mistake data are analyzed here because the other writing elements each contain only one mistake per submission. This limitation arises from the current implementation, where the LLM provides only one improvement for these writing elements. The statistics in Table 5.3 summarize the mistake frequency for grammar and vocabulary in the pre- and post-test submissions.

Writing Element	Test	Mean	Median	Std.
Grammar	Pre-test	5.68	4.00	4.01
	Post-test	3.47	2.00	2.98
Vocabulary	Pre-test	2.89	2.00	3.11
	Post-test	2.47	2.00	2.78

Table 5.3: Pre- and post-test grammar and vocabulary mistakes.

Both Table 5.3 and the plot visualized in Figure 5.3 shows a clear decrease in the average number of mistakes, with grammar mistakes dropping from 5.68 to 3.47, and vocabulary mistakes reducing from 2.89 to 2.47. This emphasizes the overall improvement in writing accuracy across these two writing elements.

Comparison of Pre- and Post-Test Scores

To assess whether the pre- and post-test scores were normally distributed, a Shapiro-Wilk test was applied to the scores for each writing element [GKC23] [DH24]. The Shapiro-Wilk test is particularly effective for small sample sizes ($n < 50$) and is used to assess whether a dataset is normally distributed by testing the null hypothesis that the data comes from a normal distribution [MPS⁺19]. It is calculated as:

$$W = \frac{\left(\sum_{i=1}^n a_i x_{(i)}\right)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

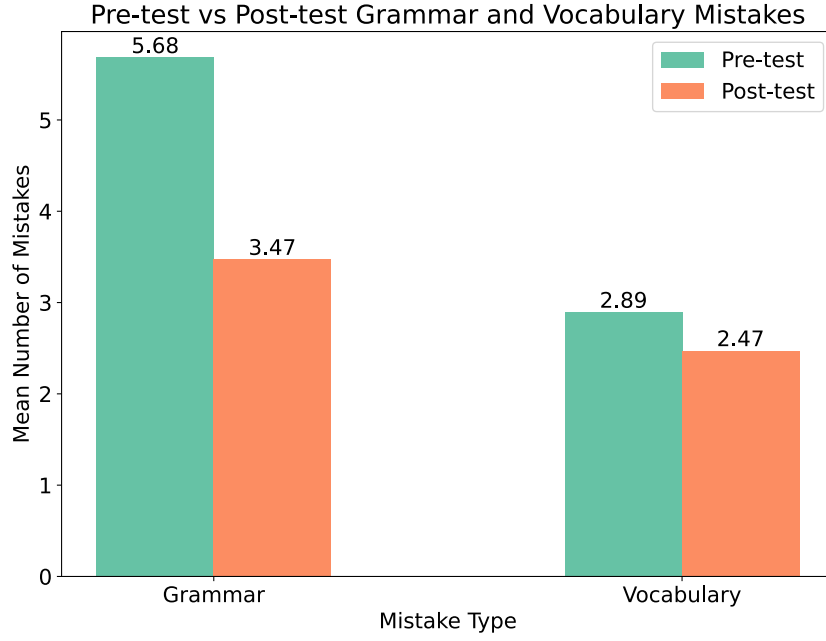


Figure 5.3: Pre-test vs post-test grammar and vocabulary mistakes.

Here, a_i are constants generated from the means, variances and covariances of the order statistics of a sample of size n from a normal distribution, $x_{(i)}$ are the ordered sample values (from smallest to largest), x_i are individual data points in the sample, and $\bar{x}$ is the mean of the sample [oST]. The results of the Shapiro-Wilk test between the pre- and post-test for each writing element are presented in Table 5.4. As the results indicate that $p > 0.05$ for all writing elements in both the pre- and post-test, it suggests that none of the data significantly deviates from a normal distribution, meaning that the pre- and post-test scores for each writing element are normally distributed. Therefore, for subsequent analysis, parametric tests can be applied to compare pre- and post-test scores.

Writing Element	Pre-/Post-Test Difference (p -value)
Grammar	0.540
Vocabulary	0.418
Organization	0.125
Coherence	0.186
Writing Style	0.539

Table 5.4: Shapiro-Wilk test for difference between pre- and post-test data.

Given the normality of the data, a paired t-test was conducted to each writing element to compare the pre- and post-test scores [PGCS19] [SS23] [DH24] [Mil17]. The test

5 Evaluation

compares the mean of the differences between paired observations (pre- and post-test scores) to zero, assessing whether the observed differences are likely due to chance [RWRW17]. The t -statistic is calculated using the following formula:

$$t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}$$

Here, $\bar{d}$ is the mean of the differences between the pre- and post-test scores, s_d is the standard deviation of the differences, and n is the number of paired observations. The p -value is calculated based on the t -statistic, which measures the ratio of the difference between the paired means to the variability in the data, with the p -value representing the probability of observing such a t -statistic under the null hypothesis of no difference. This means, that if the p -value is less than the significance level of 0.05, the null hypothesis is rejected, indicating a significant difference.

Additionally, to understand the effect size of the differences, Cohen's d was calculated providing valuable information about the practical significance of our findings. As explained by [SS23], Cohen's d is a useful measure to determine the magnitude of the difference between two groups, with values closer to 0.2 indicating small effects, around 0.5 medium effects, and 0.8 or higher indicating large effects [Bry19] [Sta]. It is calculated as follows:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_p} = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}}$$

Here, $\bar{X}_1$ and $\bar{X}_2$ are the means of the two groups, s_1^2 and s_2^2 are the variances of the two groups, and n_1 and n_2 are the sample sizes of the two groups [Wil23].

The results from Table 5.5, which presents the outcomes of the paired t -test and values of Cohens' d , indicate that the p -values for all writing elements are greater than 0.05, indicating that there is no statistically significant difference between the pre- and post-test scores across the five elements. The t -values are negative, yet they are not large enough to achieve statistical significance. However the effect sizes with grammar ($p=0.4544$) and writing style ($p=0.4434$) show the largest effects.

Writing Element	t -value	p -value	d
Grammar	-1.98	0.063	0.454
Vocabulary	-1.66	0.115	0.380
Organization	-1.57	0.133	0.361
Coherence	-1.74	0.100	0.398
Writing Style	-1.93	0.069	0.443

Table 5.5: Paired t -test results and Cohens' d .

Relation between Number of Submissions and Score Improvements

To investigate the relationship between the number of submissions and changes in test scores for each writing element, users were divided into three groups based on the independent variable, i.e. the submission rate: low (below the 25th percentile), medium (between the 25th and 75th percentile), and high (above the 75th percentile) [WHF15]. We employed one-way analysis of variance (ANOVA) to compare the test score improvements across these groups for grammar, vocabulary, coherence and writing style [KB11] [HZN21] [STP13]. ANOVA is used to determine whether there are significant differences between the means of three or more independent groups. To decide whether group means differ statistically from one another, it compares the variance within each group to the variance between the groups.

$$F = \frac{\text{Between-group variance}}{\text{Within-group variance}} = \frac{\frac{SS_{\text{between}}}{df_{\text{between}}}}{\frac{SS_{\text{within}}}{df_{\text{within}}}} = \frac{\frac{SS_{\text{between}}}{k-1}}{\frac{SS_{\text{within}}}{n-k}}$$

Here, SS_{between} is the sum of squares between the groups, SS_{within} is the sum of squares within the groups, df_{between} is the degrees of freedom between the groups, df_{within} is the degrees of freedom within the groups, k is the number of groups and n the total number of observations. Once the F -statistic is obtained, the p -value is found by comparing this F -statistic to the theoretical F -distribution with the corresponding degrees of freedom. If the p -value (in ANOVA referred to as Sig.) is below the significance threshold of 0.05, the null hypothesis is rejected, indicating that there is a statistically significant difference between the group means. Since the assumptions for conducting ANOVA were satisfied and the homogeneity of variances was verified, the three groups were compared [SW⁺89].

Additionally, a Kruskal-Wallis test was used for organization due to the non-normality of the data, according to the Shapiro-Wilk test [MPS⁺19]. The Kruskal-Wallis test is a non-parametric statistical test used for comparing the medians of three or more independent samples of equal or different sample sizes when the assumption of normally distributed data is violated. It is an alternative to one-way ANOVA. The test determines whether there are significant differences in the ranks of the data between groups, where ranks refer to the positions of data points when all observations are combined and sorted in ascending order. The test statistic H for the Kruskal-Wallis test is calculated as follows:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1)$$

Here, N is the total number of observations, k is the number of groups, R_i is the sum of ranks for the i -th group, and n_i is the number of observations in the i -th group. Once the H -statistic is obtained, the p -value is calculated by comparing this H -statistic to the chi-squared [oM] distribution with degrees of freedom equal to the number of groups minus 1. If the p -value (Sig.) is less than the significance level of 0.05, we reject the null hypothesis, pointing to a significant difference between the

5 Evaluation

group medians [KW52].

Based on the results from the ANOVA and Kruskal-Wallis tests summarized in Table 5.6, we can note that the submission count did not have a statistically significant effect on improvements in test scores for any of our writing elements. This can be evidenced by the values of Sig., which are all below 0.05.

Writing Element	Test Type	Between/ Within	<i>SS</i>	<i>df</i>	<i>F/H</i>	Sig.
Grammar	ANOVA	Between	2.821	2	$F = 1.000$	0.391
		Within	22.627	16		
Vocabulary	ANOVA	Between	1.000	2	$F = 1.087$	0.361
		Within	14.700	16		
Organization	Kruskal-Wallis	N/A			$H = 3.383$	0.184
Coherence	ANOVA	Between	4.403	2	$F = 2.264$	0.136
		Within	15.600	16		
Writing Style	ANOVA	Between	1.390	2	$F = 0.907$	0.424
		Within	12.250	16		

Table 5.6: ANOVA/Kruskal-Wallis results based on submission count.

The evaluation of WILLM’s immediate writing improvement features demonstrated improvements across all writing elements, some larger than others. Participants showed a reduction in mistakes and improved scores after receiving automated feedback from our system. Although statistical significance was not reached, the feedback provided by the system contributed to measurable enhancements in users’ writing performance.

5.2.2 Long-Term Writing Improvement

While the previous section focused on the immediate impact of feedback on users’ writing performance, this section delves into the effects of using quizzes and reviews as a long-term memory enhancement tool. We analyze quiz engagement, the relationship between quiz completion and changes in test scores, review generation count, and the relationship between review count and changes in test scores.

Quiz Data and Accuracy Rates

To begin the analysis of long-term improvement, we first examine users’ engagement with the quizzes. Table 5.7 summarizes the total number of quizzes completed, along with accuracy rates across different quiz types.

The global statistics reveal that out of 47 quizzes generated, 20 were completed, resulting in a global completion rate of 42.55%. The overall accuracy rate was 62.50%. However, when breaking down the data by quiz type, we observe significant variability in accuracy rates. For instance, the academic sentence and synonyms question types demonstrated high accuracy rates of 81.82% and 80.00%. In contrast, the revision question type had a notably lower accuracy rate of 20.00%.

Global Stats	Values
Total Quizzes Generated	47
Total Quizzes Completed	20
Total Questions Attempted	56
Total Questions Correct	35
Global Completion Rate	42.55%
Global Accuracy Rate	62.50%

Question Type	Total Questions	Correct Answers	Accuracy Rate
Academic Sentence	11	9	81.82%
Revision	10	2	20.00%
Coherence	19	14	73.68%
Synonyms	5	4	80.00%
Organization	11	6	54.55%

Table 5.7: Summary of quiz data across question types.

Relation between Quiz Completion Rate and Score Improvements

To further examine whether quiz completion rates influenced writing improvements, users were divided into three groups based on the independent variable, i.e. the quiz completion rate: low, medium, and high [WHF15]. We employed ANOVA to compare the test score improvements across these groups for each writing element [KB11] [HZN21] [STP13]. Since the assumptions for conducting ANOVA were satisfied and the homogeneity of variances was verified, the three groups were compared.

The ANOVA results summarized in Table 5.8 reveal that there are no statistically significant differences between the three groups for any writing element. For each writing element, the analysis indicates that the differences in test score improvements between users with low, medium, and high quiz completion rates were not significant. In particular, the extremely high p -value for coherence suggests that the performance across the groups in this element was nearly identical, with very little variation based on quiz completion. Overall, the F-statistics across all writing elements were low, and Sig. > 0.05 in every case confirms that none of these differences were statistically significant.

Review Data

Regarding the other long-term writing improvement feature, the following review data provides insights into how users interacted with the review feature over the course of three weeks. Table 5.9 summarizes the total number of reviews generated, along with average reviews generated per user and per session.

A total of 78 reviews were generated across all users, with an average of 4.88 reviews generated per user and an average of 1.77 reviews per session. Breaking this down by week, user activity increased significantly in the second week, with 44 reviews

Writing Element	Between/Within	<i>SS</i>	<i>df</i>	<i>F</i>	Sig.
Grammar	Between	2.388	2	0.699	0.520
	Within	17.094	10		
Vocabulary	Between	1.292	2	0.683	0.527
	Within	9.461	10		
Organization	Between	0.336	2	0.144	0.868
	Within	11.707	10		
Coherence	Between	0.030	2	0.010	0.990
	Within	15.654	10		
Writing Style	Between	0.828	2	0.414	0.672
	Within	9.990	10		

Table 5.8: ANOVA results based on quiz completion rate.

generated, compared to 7 in the first week and 25 in the third week.

Global Review Data	Values
Total Reviews Generated Across All Users	78
Average Reviews Generated per User	4.88
Average Reviews Generated per Session	1.77

Week	Number of Reviews Generated
Week 1	7
Week 2	44
Week 3	25

Table 5.9: Summary of review data across three weeks.

Relation between Review Generation Count and Score Improvements

To further assess the effect of review generation on writing performance, users were again divided into three groups based on the independent variable, i.e. the review generation count (i.e. low, medium, and high) [WHF15]. ANOVA was applied to grammar, vocabulary, coherence, and writing style as the assumptions for conducting ANOVA were satisfied and the homogeneity of variances was verified. A Kruskal-Wallis test was used for organization due to the non-normality of the data.

The ANOVA and Kruskal-Wallis results from Table 5.10 reveal that there are no statistically significant differences between the three groups for any writing element, as $p > 0.05$ in every case. For grammar, vocabulary, coherence and writing style, the analysis shows that the number of reviews generated did not significantly impact score improvements. For organization, the Kruskal-Wallis test also indicates no significant difference.

Writing Element	Test Type	Between/ Within	<i>SS</i>	<i>df</i>	<i>F/H</i>	Sig.
Grammar	ANOVA	Between	3.704	2	$F = 1.392$	0.283
		Within	17.291	13		
Vocabulary	ANOVA	Between	3.247	2	$F = 2.489$	0.122
		Within	8.479	13		
Organization	Kruskal-Wallis	N/A			$H = 0.992$	0.609
Coherence	ANOVA	Between	0.709	2	$F = 0.303$	0.744
		Within	15.229	13		
Writing Style	ANOVA	Between	1.551	2	$F = 0.971$	0.405
		Within	10.384	13		

Table 5.10: ANOVA/Kruskal-Wallis results based on review generation count.

The evaluation of WILLM’s long-term writing improvement features showed no statistically significant relationship between quiz completion or review generation, and writing performance improvements from pre-test to post-test.

5.2.3 Gamification

Having explored the impact of long-term writing improvement features, we now turn our attention to gamification elements. In this section, we measure and examine the user engagement.

Gamification Data Summary

Table 5.11 summarizes the gamification engagement data, including XP points earned, user levels, daily and weekly streaks, badges collected, quizzes completed, and correct answers. The overall statistics reveal the level of engagement users had with these gamified elements, which were designed to encourage consistent participation and motivation throughout the study.

From the results, we can observe that the average XP earned by users was 458.42, with a wide range from 0 to 2450 XP which is also confirmed by the high standard deviation of 556.59. The average level attained was 1.84, with a maximum level of 6. Daily streaks averaged 3.37, showing moderate consistency in daily engagement, though weekly streaks remained closely to 0. The average number of badges collected was 0.58, indicating low engagement with that feature. Users completed an average of 1.05 quizzes and correctly answered an average of 1.89 questions.

The above sections tackled the evaluation of the immediate and long-term writing improvement and their relation to the differences between the pre- and post-test results.

Gamification Metric	Mean	Median	Min	Max	Std Dev
XP	458.42	360.00	0	2450	556.59
Level	1.84	2.00	0	6	1.35
Daily Streaks	3.37	2.00	1	15	3.50
Weekly Streaks	0.16	0.00	0	2	0.49
Badges Count	0.58	1.00	0	2	0.59
Quizzes Completed	1.05	1.00	0	4	1.19
Correct Answers	1.89	1.00	0	10	2.57

Table 5.11: Summary of gamification data.

Additionally, the interaction of users with the tool was examined. In the following, we will continue the evaluation of WILLM by presenting the results of the SUS.

5.2.4 System Usability Scale

Even though users’ engagement with WILLM, as analyzed in the sections above, was moderate at best, the SUS results demonstrate a better perception of the system’s usability. As scores above 68 are considered above average, WILLM managed to receive an overall average score of 84.21, which was calculated using feedback from the SUS questionnaire that participants provided. According to the grading system outlined by the research of [SL16], this score places WILLM in the A+ grade range, signifying that users found the system highly usable. This high score reflects that the tool offers an intuitive, efficient, and positive user experience, aligning with the project’s goal of providing an effective academic writing improvement tool. More details on the scores of each participant can be found in the Appendix C.

5.3 Discussion

We have evaluated the effectiveness of WILLM for both immediate and long-term writing improvement, focusing on its ability to enhance academic writing skills. The evaluation specifically addressed the following research questions:

1. **RQ1:** Does automated scoring on grammar, vocabulary, organization, coherence, and writing style, along with continuous written feedback from LLMs, improve academic writing over time?
2. **RQ2:** How effective are active recall quizzes generated by LLMs in improving long-term academic writing skills?
3. **RQ3:** How effective are personalized review sessions provided by LLMs in improving long-term writing proficiency?

We evaluated RQ1 through participants’ submission rates, feedback analysis, and the results of pre- and post-test comparisons. Submission rates varied widely, with some

participants being a lot more engaged than others, affecting the amount of feedback they received. This is also reflected in the gamification data, shown by high variability in XP and daily streaks. Additionally, WILLM’s feedback mechanism worked well for grammar, vocabulary, and writing style, where feedback decreased as scores improved which indicates that lower-scoring users received more advice, as was expected. The small inconsistencies at lower scores for grammar and vocabulary may be due to the limited number of submissions contributing to the average calculation suggesting that the users who received these scores submitted shorter texts, thereby reducing the opportunity for error identification. Similarly, feedback for organization and coherence remained minimal, typically between 0 and 1 issue per submission. This is due to limitations in the prompt design, where the LLM currently identifies only a single issue for each writing element, leading to incomplete feedback. Additionally, the presence of feedback at higher scores might result from hallucinations or false positives. Furthermore, participants’ performance was generally clustered around higher intermediate scores, with the lower quartiles across all elements suggesting that even the weakest users were not necessarily scoring low. This could be attributed to the participants’ stronger academic backgrounds and English proficiency, reducing the room for significant improvement.

Pre- and post-test data suggested improvements, particularly in grammar and vocabulary, as seen in the reduction of errors and more consistent performance. Despite these practical improvements, also demonstrated by the moderate effect sizes particularly in grammar and writing style, the paired t-tests showed no statistically significant differences in scores across the five elements.

Lastly, the submission count did not have a significant effect on test score improvements, which may be explained by users relying on the tool more for immediate corrections rather than reflecting on and internalizing the feedback for long-term improvement.

In summary, while the automated scoring and continuous feedback features of WILLM did result in small improvements in some writing elements, particularly grammar and vocabulary, the system’s overall impact on academic writing enhancement was not statistically significant. However, the practical significance of these small improvements suggests that the system has the potential to aid users in honing their academic writing skills over time but refinements are needed.

Regarding RQ2, we evaluated the effectiveness of active recall quizzes generated by LLMs in improving long-term academic writing skills. The overall quiz accuracy rate was 62.50%, indicating moderate performance across users. Specific question types, such as academic sentence and synonyms, demonstrated higher accuracy rates of 81.82% and 80.00%, respectively, suggesting that users found these types easier to handle. In contrast, the organization-related questions had an accuracy of 54.55% whereas the revision questions showed a notably lower accuracy rate of 20.00%. This is further highlighted by the lower mean value of correct answers from the gamification data. These results suggest that some question types, especially revision and organization posed greater challenges for users and may require further refinement.

5 Evaluation

The statistical analysis showed no significant relationship between quiz completion rates and improvements in pre- and post-test scores for any writing element. This lack of impact is likely due to the low overall engagement with the quiz feature, as the relatively small number of completed quizzes (20) limited its potential to reinforce learning. One possible reason for this low engagement could be the UI, which made it difficult for participants to recognize when a quiz was due. The quiz availability was only briefly shown during login and was accessible through a drop-down menu that users might not have checked frequently.

While the quizzes were moderately effective for certain question types, the overall low engagement and UI design interfered with their impact on long-term writing improvement, resulting in no statistically significant impact of active recall quizzes. To better support users, the active recall quiz feature needs further improvements in question types and more intuitive quiz notifications.

For RQ3, we evaluated the effectiveness of personalized review sessions provided by LLMs in improving long-term writing proficiency. The analysis of the review data showed that users engaged moderately with the review feature, especially in week 2 indicating an increase in user activity, possibly as users became more familiar with the review system or when the feature became more integrated into their routine.

However, despite this moderate engagement, the statistical analysis showed no significant relationship between the review generation count and improvements in pre- and post-test scores. This might again be attributed to the UI that may not have clearly indicated what users needed to improve or where they had recently made progress or having provided too much information at once, resulting in overwhelming users and limiting the potential for long-term improvement. Additionally, the tips generated in the review might not have been clear enough.

In summary, while users engaged with the review feature, the feature’s effectiveness in improving writing proficiency did not result in statistically significant impact on test score improvements, suggesting also the need for further refinements.

While LLMs have made notable advancements and are able to address many tasks, the results may still be influenced by the performance of the LLMs used in WILLM. Their limitations, such as occasional hallucinations, false positives, or difficulties in handling more complex tasks like generating personalized questions, may have influenced the overall effectiveness of the tool and in turn impacted users’ progress.

While WILLM demonstrated some practical improvements in the non-native English learners’ immediate writing performance, particularly in grammar and vocabulary, it did not show statistically significant improvements across all writing elements, and its long-term impact through active recall quizzes and reflective review sessions was limited. This concludes that more development and research are needed for building a system that can effectively support long-term academic writing improvement through the methods we proposed.

5.4 Limitations

Several limitations need to be addressed that may have influenced the outcomes of our evaluation. First, the relatively small sample size ($n = 19$) limits the universality of our findings. A larger and more diverse group of participants, especially those with lower levels of writing proficiency, would provide a stronger foundation for assessing the effectiveness of WILLM, particularly among non-native English speakers.

Another limitation involves the use of GPT-3.5 for assessing the pre- and post-test writing submissions. GPT-3.5 was the default model within WILLM, thus we also selected it for the assessment of the tests. Therefore, the evaluation results may strongly depend on the capabilities of this particular language model. It is important to note that different models, such as GPT-4o or newer versions like GPT-4o mini and o1, might assess the writing tests with slightly different scores. This could result in variations in the evaluation, potentially affecting the significance of the results and leading to a different interpretation of the tool's effectiveness. Consequently, further research is needed to analyze WILLM's effectiveness with alternative LLMs to better understand the potential variability in scoring.

Lastly, self-reported data from the SUS introduces bias. While SUS provides valuable insights into user perceptions of the system's usability, self-reported measures can be subjective and may not fully reflect the actual performance or engagement with the tool.

In this chapter, we explained the methodology of our evaluation, including the pre- and post-test design and usability analysis. We presented the results, assessing the impact of WILLM's features on both immediate and long-term writing improvements, and provided a discussion of the findings along with the evaluation's limitations.

6 Conclusion and Future Work

6.1 Contributions

In this thesis, we proposed WILLM, an academic writing improvement tool aimed at assisting non-native English-speaking students in enhancing their academic writing skills. Leveraging the capabilities of LLMs, specifically GPT-3.5 and GPT-4o, we integrated immediate feedback with long-term writing improvement features into a user-friendly web application, addressing a gap in current existing academic writing tools.

Our main contribution includes the design and implementation of comprehensive feedback mechanism that provide automated scoring and detailed feedback on five critical writing elements: grammar, vocabulary, organization, coherence, and writing style. Unlike many tools that focus solely on surface-level corrections, WILLM provides deeper insights into the various aspects of academic writing. Moreover, by introducing active recall quizzes and personalized reviews, we incorporated strategies based in learning research to reinforce long-term memory retention and support writing proficiency over time. Additionally, gamification elements such as XP, levels, badges, and achievements were added to measure user interaction with the tool.

To evaluate WILLM, we conducted a three-week user study with 19 participants, utilizing a pre- and post-test design to measure improvements in writing proficiency, along with the SUS to assess user experience. The results demonstrated improvements, as evidenced by the increased consistency in mean scores: grammar improved from 6.22 to 6.76, vocabulary from 6.35 to 6.72, organization from 7.11 to 7.47, coherence from 6.68 to 7.10, and writing style from 6.07 to 6.45. Additionally, there was a reduction in errors for both grammar (from 5.68 to 3.47) and vocabulary (from 2.89 to 2.47) between the pre- and post-tests, indicating that immediate, AI-driven feedback can effectively support writing development. The paired *t*-test results showed negative *t*-values, and effect sizes from Cohen's *d* ranged between 0.2 and 0.5, indicating practical improvements too, though these tests did not achieve statistical significance ($p > 0.05$). Similarly, these improvements were not statistically significant across all writing elements. The long-term improvement features did not have a significant impact on writing proficiency, as reflected by ANOVA tests showing no significant results (Sig. > 0.05), which may be partly due to low user engagement, as quizzes and reviews were not used frequently. This outcome highlights areas for enhancement, particularly in user engagement and UI design. Despite this outcome, the high SUS score of 84.21 reflects a positive user reception and high usability.

By addressing both immediate feedback and long-term memory improvement within a single application, our work extends the current research in academic writing tools by addressing the gap that most current tools emphasize either grammar correction or language enhancement, but none combine instant feedback with mechanisms to reinforce long-term writing proficiency. This work also highlights the practical application of LLMs in educational environments, offering insights into their potential beyond immediate corrections.

In conclusion, WILLM contributes a novel solution to the challenge of improving academic writing among non-native English speakers. By combining immediate corrective feedback with strategies for long-term retention, our work offers a conceptual foundation for developing more effective educational tools that make use of the capabilities of LLM.

6.2 Improvements and Future Work

Our user study conducted involved a relatively **small sample size** over a three-week period, which limited the statistical significance of the findings. Conducting a user study with a larger participant pool and also over a longer period of time would provide more data to assess the effectiveness of WILLM. Moreover, **incorporating qualitative evaluation methods, such as interviews or focus groups, could provide deeper insights into users' experiences and perceptions of the tool.**

Since our results showed a low user engagement, it would be beneficial to **focus on refining the user interface and user experience design to make features like quizzes and reviews more accessible. Additionally, exploring gamification strategies further by introducing more dynamic and personalized rewards, challenges, or social elements could enhance motivation.**

As language models continue to evolve, future work could explore integrating newer or more specialized models that offer improved performance. Experimenting with different models for specific tasks within WILLM may enhance the quality of feedback and reduce limitations such as hallucinations. This could also lead to a better cost efficiency, as newer models sometimes provide more efficient token usage. This would allow for handling more complex tasks at a lower computational cost, potentially enabling users to make more requests within the same budget.

Furthermore, **implementing a mechanism that adjusts the difficulty of quiz questions based on the user's current progress could further personalize the learning experience.** Enhancing the feedback system to prioritize errors based on their severity could also help users focus on more critical issues in their writing. **Another direction includes the adjustment of WILLM for more experienced academic writers as they may require**

6 Conclusion and Future Work

a more sophisticated feedback and analysis of their writing. Incorporating this could make the tool valuable for a wider range of users.

To better support users in their writing process, WILLM could be expanded to allow for the submission of entire papers. Providing feedback on the organization between chapters, sections, and paragraphs would offer users a more holistic understanding of their writing structure and coherence. The integration of a chatbot feature would enable users to ask specific questions and receive personalized explanations which could help making the tool more responsive to users' needs.

Bibliography

- [A⁺21] Ariyanti Ariyanti et al. Technology-enhanced paraphrasing tool to improve efl students' writing achievement and enjoyment. *Journal of English Language Teaching and Linguistics*, 6(3):715–726, 2021.
- [AA23] Norah Almusharraf and Hind Alotaibi. An error-analysis study from an efl writing context: Human and automated essay scoring approaches. *Technology, Knowledge and Learning*, 28(3):1015–1031, 2023.
- [AAA⁺23] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [AB15] IAH Al-Badi. Academic writing difficulties of esl learners. *wei international academic conference proceedings*, 63–76, 2015.
- [AF12] Hind Al Fadda. Difficulties in academic writing: From the perspective of king saud university postgraduate students. *English Language Teaching*, 5(3):123–130, 2012.
- [Akb20] Muhammad Taufiq Akbar. Students' paraphrasing in the literature review section of research proposal. *Jambura Journal of English Teaching and Literature*, 1(1):1–15, 2020.
- [AMT19] Meilisa Sindy Astika Ariyanto, Nur Mukminatien, and Sintha Tresnadewi. *Students' and Teacher's Perceptions Towards the Implementation of ProWritingAid and Teacher Feedback*. PhD thesis, State University of Malang, 2019.
- [Ank] Anki. Powerful, intelligent flash cards. remembering things just became much easier. <https://apps.ankiweb.net/>.
- [AT18] Nasser Omer Mubarak Al-Tamimi. An investigation of academic writing errors among tertiary level students at hadhramout university: A perception study. *Journal of Applied Linguistics and Language Research*, 5(2):214–227, 2018.
- [AZA13] Silviu Apostol, Loredana Zaharescu, and Isabelle Alexe. Gamification of learning and educational games. *Elearning & Software For Education*, 2013.

Bibliography

- [BDKKM91] Robert L Bangert-Drowns, Chen-Lin C Kulik, James A Kulik, and MaryTeresa Morgan. The instructional effect of feedback in test-like events. *Review of educational research*, 61(2):213–238, 1991.
- [BKRI08] Andrew C Butler, Jeffrey D Karpicke, and Henry L Roediger III. Correcting a metacognitive error: feedback increases retention of low-confidence correct responses. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 34(4):918, 2008.
- [BL20] Algrenita Silvina Budjalemba and Listyani Listyani. Factors contributing to students difficulties in academic writing class: Students perceptions. *UC Journal: ELT, Linguistics and Literature Journal*, 1(2):135–149, 2020.
- [BMK24] Vesna Bulatović, Ivana Mirović, and Tanja Kaurin. Analyzing grammarly software for corrective feedback: Teacher’s perspective on affordances, limitations and implementation. *Focus on ELT Journal*, 6(1):74–86, 2024.
- [BMR⁺20] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Chris Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901. Curran Associates, Inc., 2020.
- [Bro96] J Brooke. Sus: A quick and dirty usability scale. *Usability Evaluation in Industry*, 1996.
- [Bry19] Christopher R Brydges. Effect size guidelines, sample size calculations, and statistical power in gerontology. *Innovation in aging*, 3(4):igz036, 2019.
- [CCBC22] Angelito Calma, Valeria Cotronei-Baird, and Austin Chia. Grammarly: An instructional intervention for writing enhancement in management education. *The International Journal of Management Education*, 20(3):100704, 2022.
- [CGA19] Wisam Chaleila and Iman Garra-Alloush. The most frequent errors in academic writing: A case of efl undergraduate arab students in israel. *English Language Teaching*, 12(7):120–138, 2019.

- [Cha19] Bo Chang. Reflection in learning. *Online learning*, 23(1):95–110, 2019.
- [CVR⁺08] Nicholas J Cepeda, Edward Vul, Doug Rohrer, John T Wixted, and Harold Pashler. Spacing effects in learning: A temporal ridgeline of optimal retention. *Psychological science*, 19(11):1095–1102, 2008.
- [CZ17] Defeng Chen and Li Zhang. Formative assessment of academic english writing for chinese efl learners. *TESOL international journal*, 12(2):47–64, 2017.
- [DCLT18] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.
- [DD17] Christo Dichev and Darina Dicheva. Gamifying education: what is known, what is believed and what remains uncertain: a critical review. *International journal of educational technology in higher education*, 14(1):1–36, 2017.
- [DD20] Hossein Dehganazadeh and Hojjat Dehganazadeh. Investigating effects of digital gamification-based language learning: A systematic review. *Journal of English Language Teaching and Learning*, 12(25):53–93, 2020.
- [Dee] DeepL. Better writing with deepl write. <https://www.deepl.com/en/write>.
- [Det11] Sebastian Deterding. Situated motivational affordances of game elements: A conceptual model. In *Gamification: Using game design elements in non-gaming contexts, a workshop at CHI*, volume 10, 2011.
- [DH24] Veri Hardinansyah Dja’far and Fitria Nur Hamidah. The effectiveness of ai technology in improving academic english writing skills in higher education. *Journal of Language and Literature Studies*, 4(3):579–593, 2024.
- [DRM⁺13] John Dunlosky, Katherine A Rawson, Elizabeth J Marsh, Mitchell J Nathan, and Daniel T Willingham. Improving students’ learning with effective learning techniques: Promising directions from cognitive and educational psychology. *Psychological Science in the Public interest*, 14(1):4–58, 2013.
- [Duo] Duolingo. The world’s best way to learn a language. <https://www.duolingo.com/>.
- [DZY17] Fei Dong, Yue Zhang, and Jie Yang. Attention-based recurrent convolutional neural network for automatic essay scoring. In Roger Levy and Lucia Specia, editors, *Proceedings of the 21st Conference on Computational Natural Language Learning (CoNLL 2017)*, pages 153–162,

- Vancouver, Canada, August 2017. Association for Computational Linguistics.
- [Eta24] Fatemeh Etaat. The effect of ai-based applications on efl writing skill development: An inquiry into integration of ai into language learning. Master's thesis, UiT Norges arktiske universitet, 2024.
- [ETS] ETS. Toefl ibt test resources. <https://www.ets.org/toefl/test-takers/ibt/about.html>.
- [FHNZC⁺23] Fiona Fui-Hoon Nah, Ruilin Zheng, Jingyuan Cai, Keng Siau, and Langtao Chen. Generative ai and chatgpt: Applications, challenges, and ai-human collaboration, 2023.
- [Fit18] Tira Nur Fitria. Error analysis found in students' writing composition of simple future tense. *Available at SSRN 3527851*, 2018.
- [Fit21] Tira Nur Fitria. Grammarly as ai-powered english writing assistant: Students' alternative for writing english. *Metathesis: Journal of English Language, Literature, and Teaching*, 5(1):65–78, 2021.
- [Fla] Flaticon. This figure has been designed using resources from flaticon.com. <https://flaticon.com>.
- [FR15] Filomena Faiella and Maria Ricciardi. Gamification and learning: a review of issues and research. *Journal of e-learning and knowledge society*, 11(3), 2015.
- [GAH15] Thikra K Ghalib and Abdulghani A Al-Hattami. Holistic versus analytic evaluation of efl writing: A case study. *English Language Teaching*, 8(7):225–236, 2015.
- [GJPK22] Shikha Gupta, Atul Jaiswal, Abinethaa Paramasivam, and Jyoti Kotecha. Academic writing challenges and supports: Perspectives of international doctoral students and their supervisors. In *Frontiers in Education*, volume 7, page 891534. Frontiers Media SA, 2022.
- [GKÇ23] Vasfiye Geçkin, Ebru Kızıldaş, and Çağatay Çınar. Assessing second-language academic writing: Ai vs. human raters. *Journal of Educational Technology and Online Learning*, 6(4):1096–1108, 2023.
- [Goo] Google. Write with ai in google docs. <https://support.google.com/docs/answer/13447609?hl=en>.
- [GR18] M Ali Ghuftron and Fathia Rosyida. The role of grammarly in assessing english as a foreign language (efl) writing. *Lingua Cultura*, 12(4):395–403, 2018.

- [Gra] Grammarly. Grammarly, free ai writing assistance. <https://www.grammarly.com>.
- [GRB19] Anita Gain, Mahabaleshwara Rao, and Shivananda K Bhat. Usage of grammarly–online grammar and spelling checker tool at the health sciences library, manipal academy of higher education, manipal: A study. *Library Philosophy and Practice*, pages 1–13, 2019.
- [HB13] Louise Horstmanshof and Sonya Brownie. A scaffolded approach to discussion board use for formative assessment of academic writing skills. *Assessment & Evaluation in Higher Education*, 38(1):61–73, 2013.
- [HF15] Michael D Hanus and Jesse Fox. Assessing the effects of gamification in the classroom: A longitudinal study on intrinsic motivation, social comparison, satisfaction, effort, and academic performance. *Computers & education*, 80:152–161, 2015.
- [HJR96] Darryl M Hunter, Richard M Jones, and Bikkar S Randhawa. The use of holistic versus analytic scoring for large-scale assessment of writing. *Canadian Journal of Program Evaluation*, 11(2):61–86, 1996.
- [HLS⁺23] Benni Ichsanda Rahman Hz, Rebecca Evelyn Laiya, Martiman Suazisiwa Sarumaha, et al. Navigating english writing proficiency tests in the era of artificial intelligence. *Journal of English Education and Teaching*, 7(3):480–498, 2023.
- [HtW] Ziwei Huang¹ and Wilco te Winkel. Automated feedback: An ai-powered tool to scale micro-level feedback for better academic writing.
- [HZN21] Yangxi Han, Shuo Zhao, and Lee-Luan Ng. How technology tools impact writing performance, lexical complexity, and perceived self-regulated learning strategies in efl academic writing: A comparative study. *Frontiers in psychology*, 12:752793, 2021.
- [IE] Team Indeed Editorial. What is academic writing? definition and 7 types to consider. <https://www.indeed.com/career-advice/career-development/academic-writing>. Accessed: 2024-08-20.
- [IEL] IELTS. Ielts scoring in detail. <https://ielts.org/organisations/ielts-for-organisations/ielts-scoring-in-detail>.
- [Ins] InstaText. Write like a native speaker. <https://instatext.io/>.
- [Ish23] FELLAH Ishak. Investigating the relationship between algerian efl learners’ use of grammarly and academic grammar writing: The case of master students at biskra university, 2023.
- [Jab18] Oqab Jabali. Students’ attitudes towards efl university writing: A case study at an-najah national university, palestine. *Heliyon*, 4(11), 2018.

Bibliography

- [KA24] Mohamed Khalifa and Mona Albadawy. Using artificial intelligence in academic writing and research: An essential productivity tool. *Computer Methods and Programs in Biomedicine Update*, page 100145, 2024.
- [KAW20] Firas Layth Khaleel, Noraidah Sahari Ashaari, and TSMT Wook. The impact of gamification on students learning engagement. *International Journal of Electrical and Computer Engineering*, 10(5):4965, 2020.
- [KB11] Jeffrey D Karpicke and Althea Bauernschmidt. Spaced retrieval: absolute spacing enhances learning regardless of relative spacing. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 37(5):1250, 2011.
- [KF22] Eka Yuni Kurniati and Rahmah Fithriani. Post-graduate students’ perceptions of quillbot utilization in english academic writing class. *Journal of English Language Teaching and Linguistics*, 7(3):437–451, 2022.
- [KK18] Elias Kyewski and Nicole C Krämer. To gamify or not to gamify? an experimental field study of the influence of badges on motivation, activity, and performance in an online learning course. *Computers & Education*, 118:25–37, 2018.
- [KMPW15] Markus Krause, Marc Mogalle, Henning Pohl, and Joseph Jay Williams. A playful game changer: Fostering student retention in online education with social gamification. In *Proceedings of the Second (2015) ACM conference on Learning@ Scale*, pages 95–102, 2015.
- [KS89] Raymond W Kulhavy and William A Stock. Feedback in written instruction: The place of response certitude. *Educational psychology review*, 1:279–308, 1989.
- [KW52] William H Kruskal and W Allen Wallis. Use of ranks in one-criterion variance analysis. *Journal of the American statistical Association*, 47(260):583–621, 1952.
- [Kwo23] Taeahn Kwon. *Interfaces for Personalized Language Learning with Generative Language Models*. PhD thesis, Columbia University, 2023.
- [KYT⁺23] Cansu Koyuturk, Mona Yavari, Emily Theophilou, Sathya Bursic, Gregor Donabauer, Alessia Telari, Alessia Testa, Raffaele Boiano, Alessandro Gabbiadini, Davinia Hernandez-Leo, et al. Developing effective educational chatbots with chatgpt prompts: Insights from preliminary tests in a case study on social media literacy. *arXiv preprint arXiv:2306.10645*, 2023.
- [LCI⁺19] Shawn Loewen, Dustin Crowther, Daniel R Isbell, Kathy Minhye Kim, Jeffrey Maloney, Zachary F Miller, and Hima Rawal. Mobile-assisted

- language learning: A duolingo case study. *ReCALL*, 31(3):293–311, 2019.
- [LCR12] Ming Liu, Rafael A Calvo, and Vasile Rus. G-asks: An intelligent automatic question generation system for academic writing support. *Dialogue & Discourse*, 3(2):101–124, 2012.
- [Lew18] James R Lewis. The system usability scale: past, present, and future. *International Journal of Human–Computer Interaction*, 34(7):577–590, 2018.
- [LM21] Linda HF Lin and Bruce Morrison. Challenges in academic writing: Perspectives of engineering faculty and l2 postgraduate research students. *English for Specific Purposes*, 63:59–70, 2021.
- [LZZ⁺23] Zehan Li, Xin Zhang, Yanzhao Zhang, Dingkun Long, Pengjun Xie, and Meishan Zhang. Towards general text embeddings with multi-stage contrastive learning, 2023.
- [MAEE24] Watheq Mansour, Salam Albatarni, Sohaila Eltanbouly, and Tamer Elsayed. Can large language models automatically score proficiency of written essays? *arXiv preprint arXiv:2403.06149*, 2024.
- [Mar24] Couchbase Product Marketing. What is vector similarity search? <https://www.couchbase.com/blog/vector-similarity-search/#:~:text=In%20vector%20similarity%20search%2C%20we,represent%20a%20word%20or%20term.,> 2024.
- [ME23] Atsushi Mizumoto and Masaki Eguchi. Exploring the potential of using an ai language model for automated essay scoring. *Research Methods in Applied Linguistics*, 2(2):100050, 2023.
- [MH21] Haggag Mohamed Haggag. Implementing artificial intelligence-powered tools in enhancing toefl-itp test takers’ reading and writing skills, 2021.
- [Mic] Microsoft. Introducing microsoft 365 copilot – your copilot for work. <https://blogs.microsoft.com/blog/2023/03/16/introducing-microsoft-365-copilot-your-copilot-for-work/>.
- [Mil17] Marine Milad. Blended learning approach: Integrating reading and writing research skills to improve academic writing. *Arab Journal of Applied Linguistics*, 3(3):23–55, 2017.
- [Moh18] Zohre Mohamadi. Comparative effect of online summative and formative assessment on efl student writing ability. *Studies in Educational Evaluation*, 59:29–40, 2018.

Bibliography

- [MPS⁺19] Prabhaker Mishra, Chandra M Pandey, Uttam Singh, Anshul Gupta, Chinmoy Sahu, and Amit Keshri. Descriptive statistics and normality tests for statistical data. *Annals of cardiac anaesthesia*, 22(1):67–72, 2019.
- [Mun17] Pilar Munday. Duolingo. gamified learning through translation. *Journal of Spanish Language Teaching*, 4(2):194–198, 2017.
- [N⁺23] Sulistya Ningrum et al. Chatgpt’s impact: The ai revolution in efl writing. *Borneo Engineering & Advanced Multidisciplinary International Journal*, 2(Special Issue (TECHON 2023)):32–37, 2023.
- [Nan23] Harini Nanayakkara. Reflection to the success of language learning. Master’s thesis, H. Nanayakkara, 2023.
- [Nex] Nexla. Vector databases: Tutorial, best practices & examples. <https://nexla.com/ai-infrastructure/vector-databases/>.
- [oM] Encyclopedia of Mathematics. Chi-squared distribution. https://encyclopediaofmath.org/index.php?title=Chi-squared_distribution.
- [Ope22] OpenAI. Introducing chatgpt, 2022.
- [oS] University of Sidney. Academic writing. <https://www.sydney.edu.au/students/writing.html>. Accessed: 2024-08-20.
- [oST] National Institute of Standards and Technology. Shapiro wilk tests. <https://www.itl.nist.gov/div898/handbook/prc/section5/prc5.htm#Pearson,%20A.%20V.,%20and%20Hartley,%20H.%200..>
- [PGCS19] Lorena Parra G and Ximena Calero S. Automated writing evaluation tools in the improvement of the writing skill. *International Journal of Instruction*, 12(2):209–226, 2019.
- [PGW23] Xiao Pu, Mingqi Gao, and Xiaojun Wan. Summarization is (almost) dead. *arXiv preprint arXiv:2309.09558*, 2023.
- [PIG] PIGAI.ORG. Score your essay. <https://en.pigai.org/>.
- [PLO⁺17] Ekaterina Pechenkina, Daniel Laurence, Grainne Oates, Daniel El-dridge, and Dan Hunter. Using a gamified mobile app to increase student engagement, retention and academic achievement. *International Journal of Educational Technology in Higher Education*, 14(1):1–12, 2017.
- [Pro] ProWritingAid. Prowritingaid: The ai-powered writing assistant. <https://prowritingaid.com/>.

- [PTE] Pearson PTE. Pte academic scoring. <https://www.pearsonpte.com/pte-academic/scoring>.
- [PWL24] James Jie Pan, Jianguo Wang, and Guoliang Li. Survey of vector database management systems. *The VLDB Journal*, pages 1–25, 2024.
- [Quia] Quillbot. Quillbot: Your complete writing solution. <https://quillbot.com/>.
- [Quib] Quizlet. How do you want to study? <https://quizlet.com/>.
- [Ref17] Lely Refnita. The effect of regular quizzes towards students’ achievement in a content-based subject. *English Language Teaching and Research*, 1(1), 2017.
- [Rei19] N Reimers. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084*, 2019.
- [RJX⁺19] Sherry Ruan, Liwei Jiang, Justin Xu, Bryce Joe-Kun Tham, Zhengneng Qiu, Yeshuang Zhu, Elizabeth L Murnane, Emma Brunskill, and James A Landay. Quizbot: A dialogue-based adaptive learning system for factual knowledge. In *Proceedings of the 2019 CHI conference on human factors in computing systems*, pages 1–13, 2019.
- [RMA23] Muhammad Ramzan, Amna Mushtaq, and Zahira Ashraf. Evacuation of difficulties and challenges for academic writing in esl learning. *University of Chitral Journal of Linguistics and Literature*, 7(I):42–49, 2023.
- [RSR⁺20] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *The Journal of Machine Learning Research*, 21(1):5485–5551, 2020.
- [RWC⁺19] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019.
- [RWRW17] Amanda Ross, Victor L Willson, Amanda Ross, and Victor L Willson. Paired samples t-test. *Basic and Advanced Statistical Tests: Writing Results Sections and Creating Tables and Figures*, pages 17–19, 2017.
- [SBA] Fatima Ezzahra SBAA. Investigating the effect of automated writing evaluation on efl learners’ writing accuracy: The case of master students at biskra university.
- [SKC⁺22] Zachari Swiecki, Hassan Khosravi, Guanliang Chen, Roberto Martinez-Maldonado, Jason M Lodge, Sandra Milligan, Neil Selwyn, and Dragan Gašević. Assessment in the age of artificial intelligence. *Computers and Education: Artificial Intelligence*, 3:100075, 2022.

Bibliography

- [SL16] Jeff Sauro and James R Lewis. *Quantifying the user experience: Practical statistics for user research*. Morgan Kaufmann, 2016.
- [Smi] Sheldon Smith. Academic writing. <https://www.eapfoundation.com/writing/what/>. Accessed: 2024-08-20.
- [SNO22] Awaz Naaman Saleem, Narmin Mohammed Noori, and Fezile Ozdamli. Gamification applications in e-learning: A literature review. *Technology, Knowledge and Learning*, 27(1):139–159, 2022.
- [SS23] Cuiping Song and Yanping Song. Enhancing academic writing skills and motivation: assessing the efficacy of chatgpt in ai-assisted language learning for efl students. *Frontiers in Psychology*, 14:1260843, 2023.
- [Sta] LibreText Statistics. Cohen’s standards for small, medium, and large effect sizes. [https://stats.libretexts.org/Bookshelves/Applied_Statistics/Business_Statistics_\(OpenStax\)/10:_Hypothesis_Testing_with_Two_Samples/10.03:_Cohen’s_Standards_for_Small_Medium_and_Large_Effect_Sizes](https://stats.libretexts.org/Bookshelves/Applied_Statistics/Business_Statistics_(OpenStax)/10:_Hypothesis_Testing_with_Two_Samples/10.03:_Cohen’s_Standards_for_Small_Medium_and_Large_Effect_Sizes).
- [STK⁺23] Mitchell Shortt, Shantanu Tilak, Irina Kuznetcova, Bethany Martens, and Babatunde Akinkuolie. Gamification in mobile-assisted language learning: A systematic review of duolingo literature from public release of 2012 to early 2020. *Computer Assisted Language Learning*, 36(3):517–554, 2023.
- [STP13] Masoud Khalili Sabet, Abdorreza Tahriri, and Parastou Gholami Pasand. The impact of peer scaffolding through process approach on efl learners’ academic writing fluency. *Theory and Practice in Language Studies*, 3(10):1893, 2013.
- [Stu01] Daniel Stufflebeam. Evaluation models. *New directions for evaluation*, 2001(89):7–98, 2001.
- [Sun09] Yu-Chih Sun. Using a two-tier test in examining taiwan graduate students’ perspectives on paraphrasing strategies. *Asia Pacific Education Review*, 10:399–408, 2009.
- [SW⁺89] Lars St, Svante Wold, et al. Analysis of variance (anova). *Chemometrics and intelligent laboratory systems*, 6(4):259–272, 1989.
- [TdCdS⁺19] Armando M Toda, Ricardo MC do Carmo, Alan P da Silva, Ig I Bitencourt, and Seiji Isotani. An approach for planning and deploying gamification concepts with social networks within educational contexts. *International Journal of Information Management*, 46:294–303, 2019.
- [TG24] KAAAN THILAKARATHNA and WMCP GODAGE. Ai and sri lankan university students: Developing a guideline for optimizing use and preventing misuse, 2024.

- [TJ23] Zhaoxuan Tan and Meng Jiang. User modeling in the era of large language models: Current research and future directions. *arXiv preprint arXiv:2312.11518*, 2023.
- [Tra23] Thi Thu Hien Tran. Ai tools in teaching and learning english academic writing skills. In *Proceedings of the AsiaCALL International Conference*, volume 4, pages 170–187, 2023.
- [Tri24] Sachin Tripathi. <https://adasci.org/a-comprehensive-guide-to-vector-databases-and-their-utilities/>. <https://adasci.org/a-comprehensive-guide-to-vector-databases-and-their-utilities/>, 2024.
- [TSB⁺24] Tamara P Tate, Jacob Steiss, Drew Bailey, Steve Graham, Youngsun Moon, Daniel Ritchie, Waverly Tseng, and Mark Warschauer. Can ai provide useful holistic essay scoring? *Computers and Education: Artificial Intelligence*, 7:100255, 2024.
- [VSP⁺17] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- [WHF15] William Wellington, David Hutchinson, and AJ Faria. The effectiveness of online quizzing as a repetitive learning tool in a marketing class: A field study of the testing effect. In *Developments in business simulation and experiential learning: Proceedings of the annual ABSEL conference*, volume 42, 2015.
- [WHN⁺22] Yongchao Wu, Aron Henriksson, Jalal Nouri, Martin Duneld, and Xiu Li. Beyond benchmarks: Spotting key topical sentences while improving automated essay scoring performance with topic-aware bert. *Electronics*, 12(1):150, 2022.
- [Wil23] DB Wilson. Practical meta-analysis effect size calculator (version 2023.11. 27), 2023.
- [Wor] Wordtune. Express yourself with confidence. <https://www.wordtune.com/>.
- [WWLL22] Yongjie Wang, Chuan Wang, Ruobing Li, and Hui Lin. On the use of bert for automated essay scoring: Joint learning of multi-scale essay representation. *arXiv preprint arXiv:2205.03835*, 2022.
- [WWS⁺22] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35:24824–24837, 2022.

Bibliography

- [XCK⁺22] Jiayi Xie, Kaiwei Cai, Li Kong, Junsheng Zhou, and Weiguang Qu. Automated essay scoring via pairwise contrastive regression. In *Proceedings of the 29th International Conference on Computational Linguistics*, pages 2724–2733, 2022.
- [YCRI22] Ann Yuan, Andy Coenen, Emily Reif, and Daphne Ippolito. Wordcraft: Story writing with large language models. In *27th International Conference on Intelligent User Interfaces, IUI '22*, page 841–852, New York, NY, USA, 2022. Association for Computing Machinery.
- [YCW⁺20] Ruosong Yang, Jiannong Cao, Zhiyuan Wen, Youzheng Wu, and Xiaodong He. Enhancing automated essay scoring performance via fine-tuning pre-trained language models with combination of regression and ranking. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1560–1569, 2020.
- [ZMSR19] Nur Aisyah Zulkifli, Mukaiyar Mukaiyar, Hermawati Syarif, and Yenni Rozimela. Challenges in assessing students’ writing for future instruction. In *Seventh International Conference on Languages and Arts (ICLA 2018)*, pages 713–722. Atlantis Press, 2019.
- [ZXC24] Xin Zhao, Jiahong Xu, and Andrew Cox. Incorporating artificial intelligence into student academic writing in higher education: the use of wordtune by chinese international students, 2024.

A Writing Rubric

A Writing Rubric

Score	Description
Grammar	
9	Uses a wide range of structures with full flexibility and accuracy; rare minor errors occur only as 'slips'.
8	Uses a wide range of structures accurately; occasional errors do not impede communication.
7	Uses a variety of complex structures; produces frequent error-free sentences; occasional inappropriacies or errors.
6	Uses a mix of simple and complex sentence forms; some errors in grammar and punctuation, but rarely impede meaning.
5	Uses only a limited range of structures; attempts complex sentences but these tend to be less accurate than simple sentences.
4	Produces frequent errors in grammar and punctuation; rarely uses subordinate clauses.
3	Frequent grammatical errors and incorrect word forms obscure meaning.
2	Continuous errors show lack of control of sentence structure.
1	Little or no evidence of sentence forms.
Vocabulary	
9	Uses a wide range of vocabulary with very natural and sophisticated control; rare minor errors occur only as 'slips'.
8	Uses a wide range of vocabulary fluently and flexibly to convey precise meanings.
7	Sufficient range of vocabulary to allow some flexibility and precision; uses less common lexical items with some awareness of style and collocation.
6	Adequate range of vocabulary for the task; attempts to use less common vocabulary but with some inaccuracy.
5	Limited range of vocabulary; may demonstrate frequent errors in word choice, spelling, and/or word formation.
4	Basic vocabulary is used repetitively; noticeable errors in word choice and formation.
3	Limited control of vocabulary and frequent errors obscure meaning.
2	Very limited range of words and expressions.
1	No control over word choice and frequent errors impede understanding.
Organization	
9	Information and ideas are logically sequenced and fully developed; clear progression throughout.
8	Logically organizes information and ideas; there is clear progression throughout.
7	Logically organizes information and ideas with clear progression; uses a range of cohesive devices appropriately.

Score	Description
6	Arranges information and ideas coherently with some progression; cohesive devices may be mechanical.
5	Presents information with some organization but there may be a lack of overall progression.
4	Limited organization; may not logically sequence ideas.
3	Ideas are not arranged coherently; limited use of cohesive devices.
2	Very little control of organizational features; ideas are not logically arranged.
1	Fails to communicate any organized ideas.
Coherence	
9	Cohesive devices are used in such a way that it attracts no attention; ideas are fully developed and well-supported.
8	Cohesive devices used effectively; ideas are well-developed and supported.
7	Uses cohesive devices effectively; presents clear progression of ideas.
6	Uses cohesive devices, though cohesion within and/or between sentences may be faulty or mechanical.
5	Uses some cohesive devices but these may be inaccurate or repetitive.
4	Relationships between ideas are unclear; limited use of cohesive devices.
3	Lack of coherence; ideas are difficult to follow.
2	Minimal use of cohesive devices; ideas are not logically connected.
1	No coherent ideas are presented.
Writing Style	
9	Engages the reader with an original and compelling style; tone and register are entirely appropriate for academic writing.
8	Writing is engaging and the tone is appropriate for academic writing; minor errors in style do not detract from the overall impact.
7	Demonstrates some complexity and flexibility in style; tone and register are appropriate for academic writing.
6	Shows some variety in style and an attempt to engage the reader; tone may be uneven but generally appropriate for academic writing.
5	Writing is somewhat engaging; tone and register may be inconsistent but some attempt at academic style is evident.
4	Limited engagement with the reader; writing style may be flat and uninteresting.
3	Writing style is simplistic and fails to engage the reader; tone is often inappropriate for academic writing.
2	Very limited writing style; fails to engage the reader.

A Writing Rubric

1	No attempt to engage the reader; writing style is inappropriate for academic writing.
---	---

Table A.1: Writing Rubric Criteria and Descriptions

B Gamification

Category	Milestone/Badge	Criteria/XP Rewards
Standard Rewards	Daily Login	Log in daily (50 XP)
	Correct Answer	Give a correct answer (10 XP)
	Max Score Bonus	Get a perfect quiz score (30 XP)
Achievements	Quizzes Completed	
	Stage 1	Complete 1 quiz (100 XP)
	Stage 2	Complete 3 quizzes (200 XP)
	Stage 3	Complete 5 quizzes (300 XP)
	Correct Answers	
	Stage 1	Answer 5 questions correctly (100 XP)
	Stage 2	Answer 10 questions correctly (200 XP)
	Stage 3	Answer 20 questions correctly (300 XP)
	Weekly Streaks	
	Stage 1	Achieve 1 weekly streak (100 XP)
	Stage 2	Achieve 2 weekly streaks (200 XP)
	Stage 3	Achieve 3 weekly streaks (300 XP)
	Stage 4	Achieve 4 weekly streaks (400 XP)
	Consecutive Days	
	Stage 1	Log in daily for 3 consecutive days (50 XP)
	Stage 2	Log in daily for 7 consecutive days (150 XP)
	Stage 3	Log in daily for 14 consecutive days (300 XP)
	Stage 4	Log in daily for 21 consecutive days (500 XP)
Badges	Rookie	Complete 1 quiz
	Pro	Complete 3 quizzes, log in for 5 consecutive days, answer 10 questions correctly
	Leader	Complete 4 quizzes, answer 15 questions correctly, achieve 1 weekly streak
	Guru	Complete 5 quizzes, answer 20 questions correctly

Table B.1: Standard Rewards, Achievements and Badges with Criteria and XP Rewards inside WILLM

C System Usability Scale and Detailed Results

C System Usability Scale and Detailed Results

Question Nr.	Question
1	I think that I would like to use this system frequently.
2	I found the system unnecessarily complex.
3	I thought the system was easy to use.
4	I think that I would need the support of a technical person to be able to use this system.
5	I found the various functions in this system were well integrated.
6	I thought there was too much inconsistency in this system.
7	I would imagine that most people would learn to use this system very quickly.
8	I found the system very cumbersome to use.
9	I felt very confident using the system.
10	I needed to learn a lot of things before I could get going with this system.

Table C.1: The questions from the SUS questionnaire [Lew18]

User Nr.	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	SUS Score
1	5	2	5	1	4	1	4	1	4	1	90.0
2	2	1	5	1	4	3	4	1	5	1	82.5
3	4	3	5	2	4	2	5	1	4	2	80.0
4	2	2	3	1	3	2	4	2	3	1	67.5
5	4	2	4	1	3	4	5	1	4	1	77.5
6	4	2	4	1	4	2	4	2	4	2	77.5
7	4	2	5	1	5	1	5	1	5	1	95.0
8	4	1	5	3	5	1	5	1	5	1	92.5
9	4	2	4	1	3	2	5	2	4	1	80.0
10	4	2	5	1	5	2	5	1	5	1	92.5
11	3	1	4	1	5	1	5	2	5	1	90.0
12	3	1	5	2	4	3	5	2	4	1	80.0
13	5	1	5	1	4	1	5	1	5	1	97.5
14	3	1	4	1	4	2	5	1	3	1	82.5
15	4	2	4	1	5	2	5	2	5	1	87.5
16	3	1	4	1	4	3	4	1	2	1	75.0
17	4	1	4	1	4	3	5	2	4	1	82.5
18	4	2	4	2	3	3	4	2	4	2	70.0
19	5	1	5	1	5	1	5	1	5	1	100.0

Table C.2: SUS Scores for Each User Across 10 Questions